

Flyback 400V/12V design

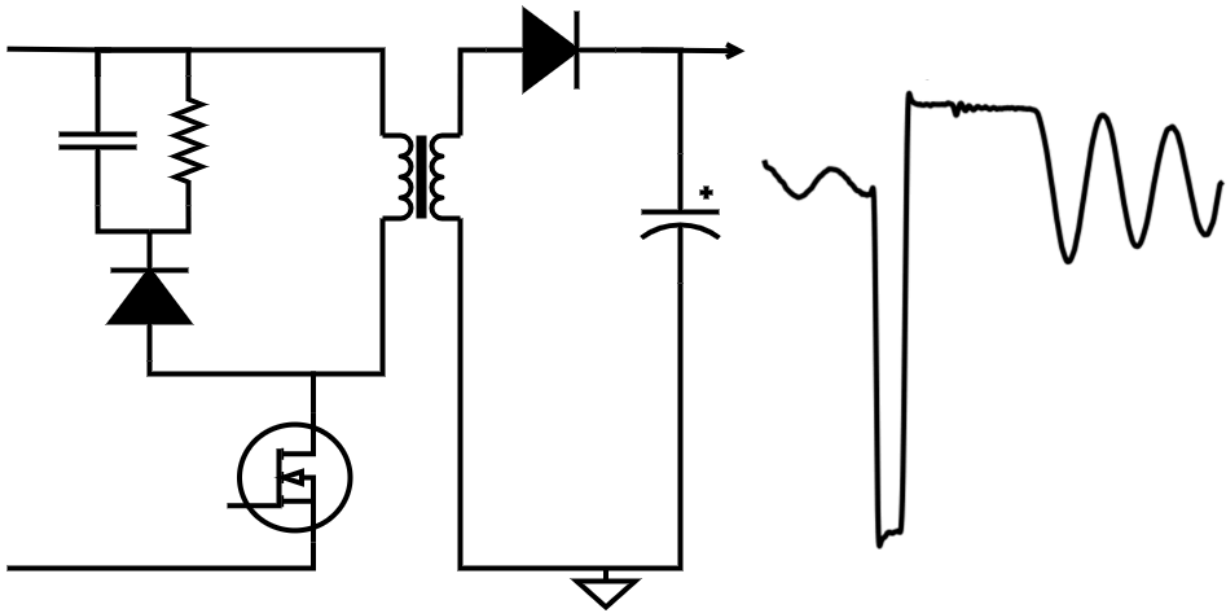


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Introduction

Inputs

$$V_{in_{nom}} = 380 \text{ (v)}$$

$$V_{in_{max}} = 400 \text{ (v)}$$

$$V_{in_{min}} = 360 \text{ (v)}$$

$$V_{out_{nom}} = 12 \text{ (v)}$$

$$Power_{out} = 3.400 \text{ (w)}$$

$$\delta V_{out} = 0.010 \text{ (1 percent)}$$

Simplified overall schematic

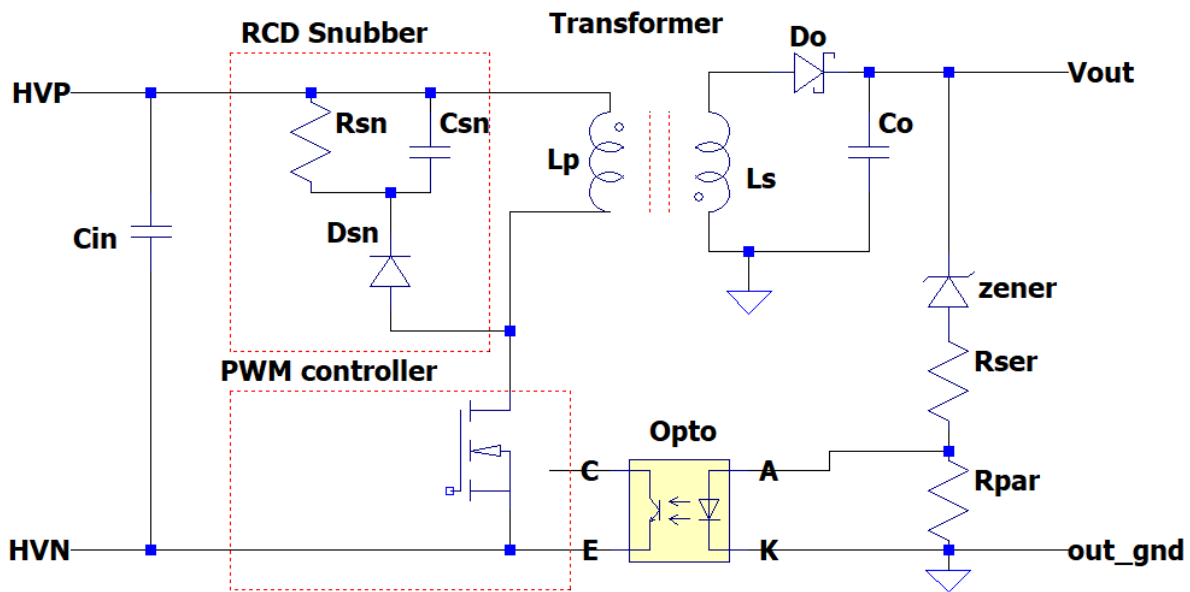


Figure 1: Simplified overall schematic

The PWM controller

Introduction of the TNY2XX family

We will use a **TNY2xx family** controller

Why:

- This controller has an **integrated MOSFET**
- It requires a **minimum BOM**
- It can operate **without an auxiliary winding**
- It provides integrated frequency jitter (**spread spectrum** to reduce EMI perturbation)

How this controller works ():

- This controller works with **max-current / max-duty-cycle**
- It has an **internal clock** that fixes the maximum switching frequency

- example: **132 kHz**
- In each cycle, it checks the **feedback pin**
 - if the feedback pin is activated → **skip (escape) the cycle**
 - otherwise → **turn on the MOSFET**
- The MOSFET turns off when **one of the following conditions is met:**
 - the **primary current** reaches the maximum value
 - example: **250 mA**
 - or the **duty cycle** reaches the maximum allowed value
 - example: **62%**
- This controller has an **internal 5.8 V generator**
 - used for **self-powering**
 - no external auxiliary supply is required

For more information please check the datasheet below [\[1\]](#).

In this design, we will show the **TNY-264** in open frame configuration

- Breakdown Voltage = 700
- Power = 9W
- Frequency = 132kHz with +/-8kHz of jitter
- $R_{ds}(ON) = 28\Omega$ @ 25°C and 42Ω @ 100°C (typ), 48Ω in the worst case
- $I_{lim} = 250\text{mA}$ (typ)
- Max duty cycle = 65% (typ)

Let's estimate the theoretical maximum power that this controller can extract

$$I_{lim} = 0.250 \text{ (A)}$$

$$F_{req} = 132000.000 \text{ (Hz)}$$

$$F_{req_{max}} = 140000.000 \text{ (Hz)}$$

$$DC_{max} = 0.650$$

$$V_{in_{max}} = 400 \text{ (V)}$$

$$d_i = I_{lim} - 0 = 0.250 - 0 = 0.250 \text{ (A)}$$

$$\begin{aligned}
 dt &= DC_{max} \cdot \left(\frac{1}{\text{Freq}} \right) \\
 &= 6.500 \times 10^{-1} \cdot \left(\frac{1}{1.320 \times 10^5} \right) \\
 &= 4.924 \times 10^{-6} \text{ (s)}
 \end{aligned}$$

$$\begin{aligned}
 L &= \frac{V_{in_{max}}}{\frac{di}{dt}} \\
 &= \frac{400}{\frac{2.500 \times 10^{-1}}{4.924 \times 10^{-6}}} \\
 &= 7.879 \times 10^{-3} \text{ (H because } v = L \, di/dt)
 \end{aligned}$$

$$\begin{aligned}
 L_{uH} &= L \cdot 1 \times 10^6 \\
 &= 7.879 \times 10^{-3} \cdot 1 \times 10^6 \\
 &= 7.879 \times 10^3 \text{ (}\mu\text{H)}
 \end{aligned}$$

$$\begin{aligned}
 E_{cycle} &= 1 \times 10^6 \cdot \frac{1}{2} \cdot L \cdot (I_{lim})^2 \\
 &= 1 \times 10^6 \cdot \frac{1}{2} \cdot 7.879 \times 10^{-3} \cdot (2.500 \times 10^{-1})^2 \\
 &= 2.462 \times 10^2 \text{ (}\mu\text{J)}
 \end{aligned}$$

$$\begin{aligned}
 P_{inp} &= \text{Freq} \cdot E_{cycle} \cdot 1 \times 10^{-6} \\
 &= 132000.000 \cdot 246.212 \cdot 1 \times 10^{-6} \\
 &= 32.500 \text{ (W)}
 \end{aligned}$$

$$\text{Efficiency} = 0.800 \text{ (hypothesis)}$$

$$\begin{aligned}
 P_{out} &= P_{inp} \cdot \text{Efficiency} \\
 &= 32.500 \cdot 0.800 \\
 &= 26.000 \text{ (W > 9W of the datasheet)}
 \end{aligned}$$

Let's estimate the thermal discipation in this condition (only conduction losses)

$$R_{ds_{max}} = 48$$

$$\begin{aligned} E_{rds_{cycle}} &= \left(\frac{1}{3}\right) \cdot R_{ds_{max}} \cdot (I_{lim})^2 \cdot dt \cdot 1 \times 10^6 \\ &= \left(\frac{1}{3}\right) \cdot 48 \cdot (0.250)^2 \cdot 0.000 \cdot 1 \times 10^6 \\ &= 4.924 \text{ (uJ (current is a slope from 0 to } I_{lim})) \end{aligned}$$

$$\begin{aligned} P_{rds} &= \text{Freq} \cdot E_{rds_{cycle}} \cdot 1 \times 10^{-6} \\ &= 132000.000 \cdot 4.924 \cdot 1 \times 10^{-6} \\ &= 0.650 \text{ (W)} \end{aligned}$$

$$R_{th_{JA}} = 45 \text{ (} ^\circ\text{C/W see the datasheet)}$$

$$\Delta T = 29.250 \text{ (} ^\circ\text{C)}$$

- The ΔT in this condition is relatively low, because in this quick calculation we assume that the only loss inside the IC is the ohmic loss caused by R_{DS_ON}
- In reality, there are additional losses, such as switching losses (especially at 132 kHz) and the internal power consumption of the controller.

In our design, the maximum output power will be 4 W, so we respect the datasheet.

Simulation for Understanding TNY-264, TNY-280 ...

To clearly understand how this controller works, I suggest the following step-by-step simulation, beginning with the simplest configuration and gradually introducing more advanced features. The goal is to improve conceptual understanding, not to provide a fully rigorous or exact model of the controller.

You can find all these simplified models in the [link](#).

Level 1 :

At this level, the PWM controller model includes only a master frequency clock (set to 132.45 kHz in the example below) and a current-limit reset.

Even with this simplified model, the controller is able to perform primary peak-current control.

See the purple current curve in the simulation shown below.

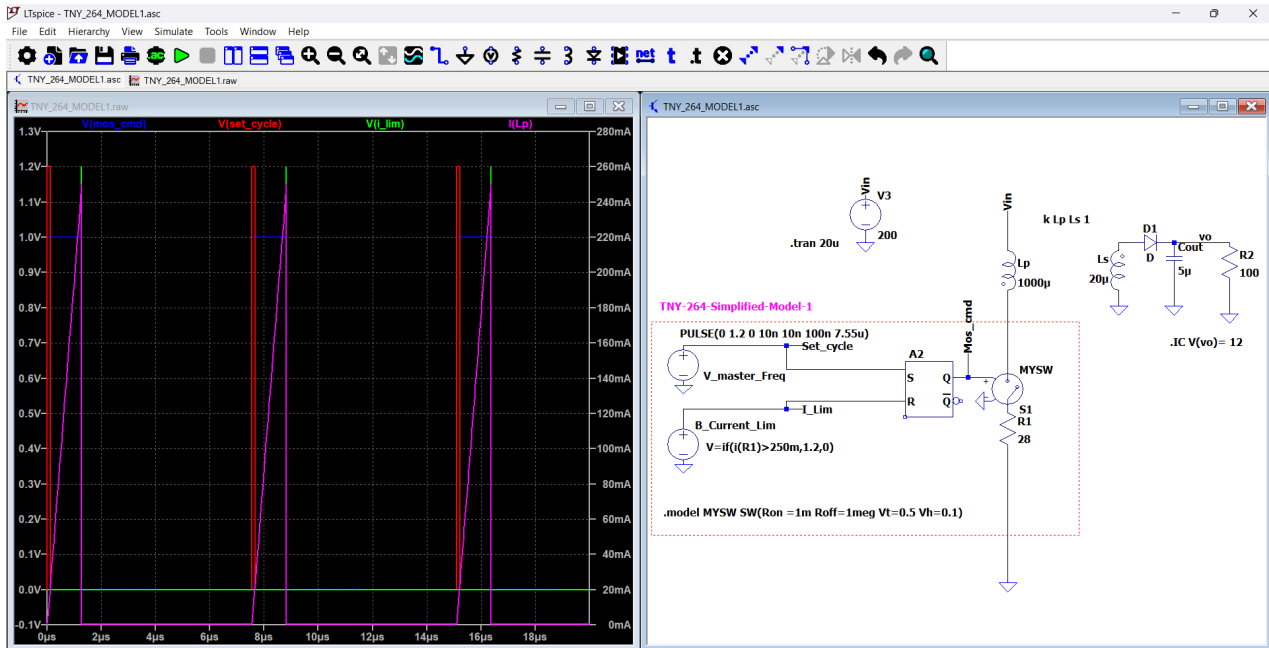


Figure 2: TNY2xx modeling: Level 1, reset triggered by the max-current.

Level 2 :

At this stage, the maximum duty-cycle reset control is added. The controller resets whenever one of the following conditions occurs:

- The current reaches its maximum target value.
- The duty cycle exceeds the controller's maximum allowed value (approximately 63%).

In the first figure below, the reset is always triggered by the maximum current.

For the second figure, the reset is triggered by the maximum duty cycle.

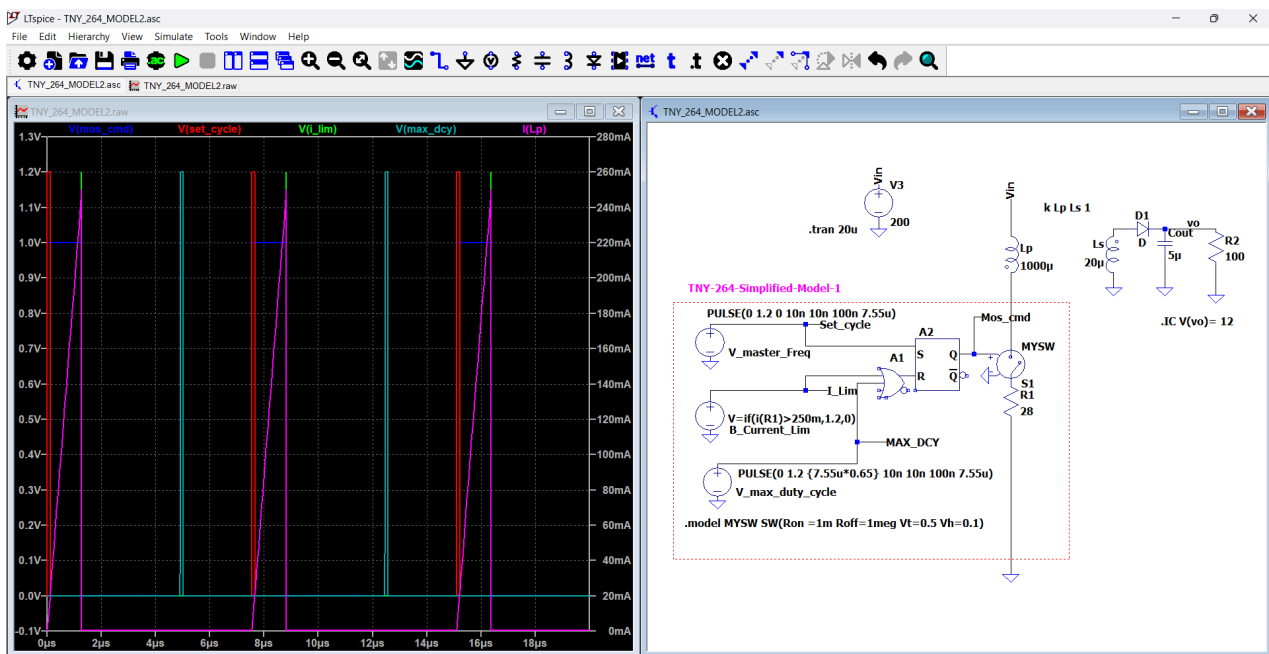


Figure 3: TNY2xx modeling: Level 2, reset triggered by the max-current.

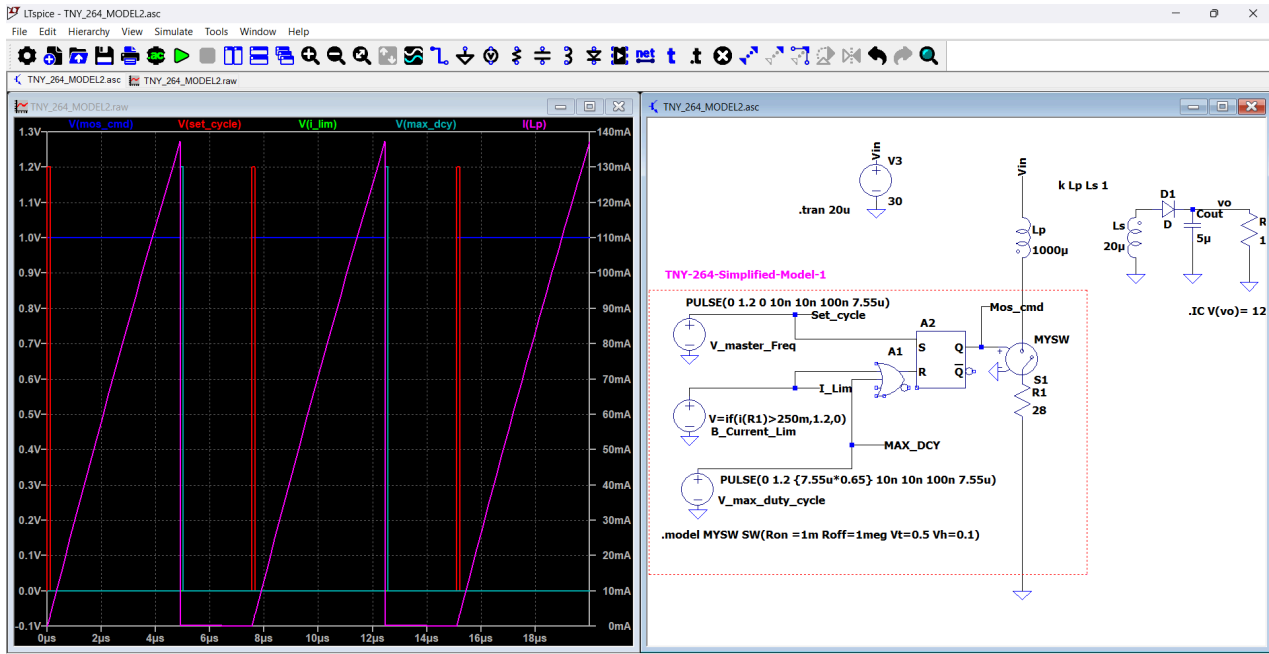


Figure 4: TNY2xx modeling: Level 2, reset triggered by the max-duty cycle.

Level 3 :

At this level, we introduce the feedback control. The feedback is evaluated at the start of each cycle, and if it is active, the current cycle is skipped. This results in fewer cycles, reduced current, and lower voltage until the feedback is deactivated, after which normal operation resumes.

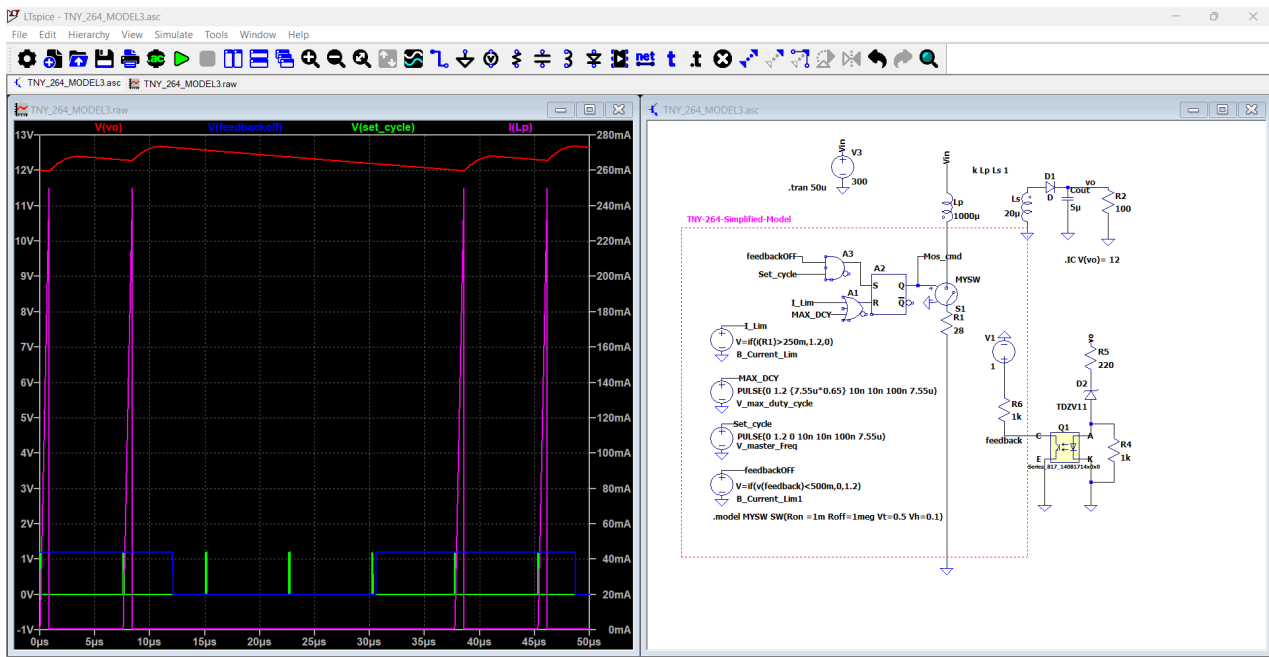


Figure 5: 2D drawing of core TDK ELP 22/6/16 with I 22/2.5/16 N87

The transformer design

let's start with this TDK planner small core **ELP 22/6/16 with I 22/2.5/16 N87**

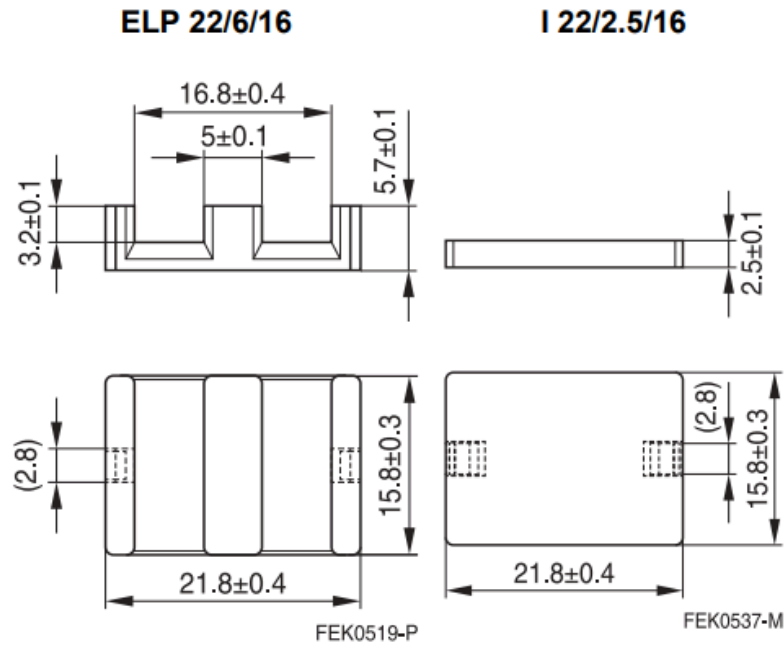


Figure 6: 2D drawing of core TDK ELP 22/6/16 with I 22/2.5/16 N87

$$B_s = 0.300 \text{ (Saturation Flux Density (T))}$$

$$A_{e_{mm2}} = 78.500 \text{ (Effective Cross Section Area (mm}^2\text{))}$$

$$l_{e_{mm}} = 26.100 \text{ (Magnetic Path Length (mm))}$$

$$\mu_i = 1360 \text{ (Initial Permeability (N87 material))}$$

$$AL = 5200 \text{ (nH :type)}$$

For more information please check the datasheet below [\[1\]](#).

$$\begin{aligned} \mu_0 &= 4 \cdot \pi \cdot 10^{-7} \\ &= 4 \cdot \pi \cdot 10^{-7} \\ &= 1.257 \times 10^{-6} \text{ (Permeability of Free Space)} \end{aligned}$$

$$\begin{aligned} A_e &= A_{e_{mm2}} \cdot 10^{-6} \\ &= 78.500 \times 10^{-6} \\ &= 7.850 \times 10^{-5} \text{ (m}^2\text{)} \end{aligned}$$

$$\begin{aligned} l_e &= l_{e_{mm}} \\ &= 26.1 \times 10^{-3} \\ &= 2.610 \times 10^{-2} \\ &= 26.1 \times 10^{-3} \\ &= 2.610 \times 10^{-2} \text{ (m)} \end{aligned}$$

One example calculation

For a generic flyback converter, the calculation is like this:

$$V_d = 0.400 \text{ (Diode Forward Drop in V)} \quad N_p = 50 \quad N_s = 10$$

$$\begin{aligned}
 n &= \frac{N_p}{N_s} \\
 &= \frac{50}{10} \\
 &= 5.000 \times 10^0
 \end{aligned}$$

$$\begin{aligned}
 V_{refl} &= n \cdot (V_{out_{nom}} + V_d) \\
 &= 5.000 \times 10^0 \cdot (12 + 4.000 \times 10^{-1}) \\
 &= 6.200 \times 10^1
 \end{aligned}$$

$$\begin{aligned}
 D &= \frac{V_{refl}}{V_{in_{max}} + V_{refl}} \\
 &= \frac{6.200 \times 10^1}{400 + 6.200 \times 10^1} \\
 &= 1.342 \times 10^{-1}
 \end{aligned}$$

$$\begin{aligned}
 T_{on} &= \frac{D}{F_{req}} \\
 &= \frac{1.342 \times 10^{-1}}{1.320 \times 10^5} \\
 &= 1.017 \times 10^{-6}
 \end{aligned}$$

$$\begin{aligned}
 B_{peak} &= \frac{V_{in_{max}} \cdot T_{on}}{N_p \cdot A_e} \\
 &= \frac{400 \cdot 1.017 \times 10^{-6}}{50 \cdot 7.850 \times 10^{-5}} \\
 &= 1.036 \times 10^{-1}
 \end{aligned}$$

$$\begin{aligned}
 L_m &= A_L \cdot (N_p)^2 \cdot 1 \times 10^{-9} \\
 &= 5200 \cdot (50)^2 \cdot 1 \times 10^{-9} \\
 &= 1.300 \times 10^{-2} \quad (\text{convert nH to Henrys})
 \end{aligned}$$

$$\begin{aligned}
 I_{p_{peak}} &= \frac{V_{in_{max}} \cdot T_{on}}{L_m} \\
 &= \frac{400 \cdot 1.017 \times 10^{-6}}{1.300 \times 10^{-2}} \\
 &= 3.128 \times 10^{-2}
 \end{aligned}$$

$$\begin{aligned}
 I_{s_{peak}} &= I_{p_{peak}} \cdot n \\
 &= 3.128 \times 10^{-2} \cdot 5.000 \times 10^0 \\
 &= 1.564 \times 10^{-1}
 \end{aligned}$$

$$\text{Effec} = 8.000 \times 10^{-1}$$

$$\begin{aligned} P_s &= 0.5 \cdot L_m \cdot (I_{\text{peak}})^2 \cdot \text{Freq} \cdot \text{Effec} \\ &= 0.5 \cdot 1.300 \times 10^{-2} \cdot (3.128 \times 10^{-2})^2 \cdot 1.320 \times 10^5 \cdot 8.000 \times 10^{-1} \\ &= 6.717 \times 10^{-1} \end{aligned}$$

$$\begin{aligned} \text{Margin} &= \frac{B_s - B_{\text{peak}}}{B_s} \\ &= \frac{3.000 \times 10^{-1} - 1.036 \times 10^{-1}}{3.000 \times 10^{-1}} \\ &= 6.546 \times 10^{-1} \end{aligned}$$

$$\begin{aligned} \text{flux} &= B_s \cdot A_e \\ &= 3.000 \times 10^{-1} \cdot 7.850 \times 10^{-5} \\ &= 2.355 \times 10^{-5} \end{aligned}$$

$$\begin{aligned} \lambda &= \text{flux} \cdot N_p \\ &= 2.355 \times 10^{-5} \cdot 50 \\ &= 1.177 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} I_{\text{sat}} &= \lambda \cdot \frac{N_p}{L_m} \\ &= 1.177 \times 10^{-3} \cdot \frac{50}{1.300 \times 10^{-2}} \\ &= 4.529 \times 10^0 \text{ (please keep in mind the margin)} \end{aligned}$$

But for our **max-current max-duty-cycle TNY-264** the calculation is below

In first time let's focus on the primary side

- Initial time step:

$$dt = DC_{\text{max}} \cdot \frac{1}{\text{Freq}}$$

- Calculate maximum DC current:

$$I_{\text{dcmax}} = \frac{V_{\text{in_nom}} \cdot dt}{L_m}$$

- Conditional calculation:

- If $I_{\text{dcmax}} < I_{\text{lim}}$:

$$I_{\text{peak}} = I_{\text{dcmax}}$$

$$D = DC_{\text{max}}$$

- Else:

$$I_{\text{peak}} = I_{\text{lim}}$$

$$dt = \frac{L_m \cdot I_{lim}}{V_{in_nom}}$$

$$D = \frac{dt}{Freq}$$

$$L_m = AL \cdot (N_p)^2 \cdot 1 \times 10^{-9} = 5200 \cdot (50)^2 \cdot 1 \times 10^{-9} = 1.300 \times 10^{-2} \text{ (H)}$$

$$dt = DC_{max} \cdot \left(\frac{1}{Freq} \right) = 6.500 \times 10^{-1} \cdot \left(\frac{1}{1.320 \times 10^5} \right) = 4.924 \times 10^{-6} \text{ (s)}$$

$$I_{dcmax} = V_{in_nom} \cdot \frac{dt}{L_m} = 380 \cdot \frac{4.924 \times 10^{-6}}{1.300 \times 10^{-2}} = 1.439 \times 10^{-1} \text{ (v = L di/dt)}$$

$$I_{peak} = 1.439 \times 10^{-1} \text{ (since Idcmax < Ilim)}$$

$$B_{peak} = V_{in_nom} \cdot \frac{dt}{N_p \cdot A_e} = 380 \cdot \frac{0.000}{50 \cdot 0.000} = 0.477 \text{ (> Bsat = 0.3 Wb)}$$

$$\text{Margin} = \frac{B_s - B_{peak}}{B_s} = \frac{0.300 - 0.477}{0.300} = -0.589 \text{ (NOK)}$$

The primary stored power

$$\text{Power}_{prim} = Freq \cdot \left(\frac{1}{2} \right) \cdot L_m \cdot (I_{peak})^2 = 132000.000 \cdot \left(\frac{1}{2} \right) \cdot 0.013 \cdot (0.144)^2 = 17.777 \text{ (W)}$$

Below a table of some Np value and corresponding Bpeak, Margin, Duty cycle (D), ...

	Np	D	Ipeak	Lm_uH	Power_prim	Bpeak	Margin %
0	10	0.045	0.250000	520.0	2.145000	0.166	44.798301
1	20	0.181	0.250000	2080.0	8.580000	0.331	-10.403397
2	30	0.406	0.250000	4680.0	19.305000	0.497	-65.605096
3	40	0.650	0.224905	8320.0	27.775805	0.596	-98.642476
4	50	0.650	0.143939	13000.0	17.776515	0.477	-58.913981
5	60	0.650	0.099958	18720.0	12.344802	0.397	-32.428317
6	70	0.650	0.073438	25480.0	9.069651	0.341	-13.509986
7	80	0.650	0.056226	33280.0	6.943951	0.298	0.678762
8	90	0.650	0.044426	42120.0	5.486579	0.265	11.714455
9	100	0.650	0.035985	52000.0	4.444129	0.238	20.543010

Filter with dutyCycle > 10%

	Np	D	Ipeak	Lm_uH	Power_prim	Bpeak	Margin %
1	20	0.181	0.250000	2080.0	8.580000	0.331	-10.403397
2	30	0.406	0.250000	4680.0	19.305000	0.497	-65.605096
3	40	0.650	0.224905	8320.0	27.775805	0.596	-98.642476
4	50	0.650	0.143939	13000.0	17.776515	0.477	-58.913981
5	60	0.650	0.099958	18720.0	12.344802	0.397	-32.428317
6	70	0.650	0.073438	25480.0	9.069651	0.341	-13.509986
7	80	0.650	0.056226	33280.0	6.943951	0.298	0.678762
8	90	0.650	0.044426	42120.0	5.486579	0.265	11.714455
9	100	0.650	0.035985	52000.0	4.444129	0.238	20.543010

So we can see that it's difficult to meet our requirement without adding an air gap and while keeping Np reasonable, so let's try adding an air gap.

Below is the calculation of the air gap using the TDK formula.

$$AL = 2400 \text{ (nH : a chosen value)}$$

$$K1 = 134 \text{ (see the core datasheet)}$$

$$K2 = -0.806 \text{ (see the core datasheet)}$$

$$\text{air}_{gap} = \left(\frac{AL}{K1} \right)^{\left(\frac{1}{K2} \right)} = \left(\frac{2400}{134} \right)^{\left(\frac{1}{-0.806} \right)} = 0.028 \text{ (AL in nH and gap in mm)}$$

For airgap calculation, see [\[4\]](#).

	Np	D	Ipeak	Lm_uH	Power_prim	Bpeak	Margin %
0	16	0.053	0.25	614.4	2.5344	0.122	59.235669
1	18	0.068	0.25	777.6	3.2076	0.138	54.140127
2	20	0.083	0.25	960.0	3.9600	0.153	49.044586
3	22	0.101	0.25	1161.6	4.7916	0.168	43.949045
4	24	0.120	0.25	1382.4	5.7024	0.183	38.853503
5	26	0.141	0.25	1622.4	6.6924	0.199	33.757962
6	28	0.163	0.25	1881.6	7.7616	0.214	28.662420
7	30	0.188	0.25	2160.0	8.9100	0.229	23.566879

Let's start with Np = 26 turns; it's a good compromise between primary power and Bpeak.

Let's chose an Ns

To avoid saturation of the core, the magnetic flux (B) must return to 0 in each cycle. This means that the magnetizing inductance must discharge completely into the secondary before the end of the cycle. Therefore, the final magnetizing current must be 0A.

Below an example with $N_s = 2$

- D : The duty cycle corresponding to T_{on} on the primary mosfet
- $D_{secondary}$: The duty cycle corresponding to T_{on} on the secondary diode
- $D_{conduction}$: The sum of the above duty cycles must be less than one

$$N_p = 26$$

$$N_s = 2 \text{ (example)}$$

$$n = \frac{N_p}{N_s} = \frac{26}{2} = 13.000$$

$$D = 0.141 \text{ (see above table for } N_p = 26)$$

$$I_{peak} = 0.250 \text{ (A see above table for } N_p = 26)$$

$$L_{m_{uH}} = 1622.400 \text{ (uH see above table for } N_p = 26)$$

$$I_{sec_{peak}} = n \cdot I_{peak} = 13.000 \cdot 0.250 = 3.250$$

$$dI_{sec} = I_{sec_{peak}} - 0 = 3.250 - 0 = 3.250$$

$$V_o = V_{out_{nom}} + V_d = 12 + 0.400 = 12.400 \text{ (V)}$$

$$\begin{aligned} L_{ms_{uH}} &= \frac{L_{m_{uH}}}{(n)^2} \\ &= \frac{1622.400}{(13.000)^2} \\ &= 9.600 \text{ (uH secondary magnetic inductance)} \end{aligned}$$

$$\begin{aligned} dtSec_{us} &= L_{ms_{uH}} \cdot \frac{dI_{sec}}{V_o} \\ &= 9.600 \cdot \frac{3.250}{12.400} \\ &= 2.516 \text{ (us reminder } v=Ldi/dt) \end{aligned}$$

$$\begin{aligned} D_{secondary} &= dtSec_{us} \cdot 1 \times 10^{-6} \cdot Freq \\ &= 2.516 \cdot 1 \times 10^{-6} \cdot 132000.000 \\ &= 0.332 \end{aligned}$$

$$\begin{aligned}
 D_{conduction} &= D + D_{secondary} \\
 &= 0.141 + 0.332 \\
 &= 0.473 \text{ (must be } < 1)
 \end{aligned}$$

$$\begin{aligned}
 V_{refl} &= (V_{out_{nom}} + V_d) \cdot n \\
 &= (12 + 0.400) \cdot 13.000 \\
 &= 161.200 \text{ (V : This voltage stresses the primary MOSFET)}
 \end{aligned}$$

Let's repeat the same calculation for different N_s :

	N_s	I_{sec_peak}	L_{ms_uH}	$dtSec_us$	$D_secondary$	$D_conduction$	V_{refl}
0	2	3.250000	9.6	2.516129	0.332129	0.473129	161.200000
1	3	2.166667	21.6	3.774194	0.498194	0.639194	107.466667
2	4	1.625000	38.4	5.032258	0.664258	0.805258	80.600000
3	5	1.300000	60.0	6.290323	0.830323	0.971323	64.480000
4	6	1.083333	86.4	7.548387	0.996387	1.137387	53.733333
5	7	0.928571	117.6	8.806452	1.162452	1.303452	46.057143
6	8	0.812500	153.6	10.064516	1.328516	1.469516	40.300000

The idea is to pick the highest N_s that ensures the secondary cycle discharges completely, in order to minimize the secondary peak current.

With **$N_s = 4$** , the full discharge of the magnetic tank is ensured: a good compromise between the $D_{conduction}$ condition and V_{refl} .

So we will start with **$N_p = 26$ and $N_s = 4$ and an $AL = 2400$ nH that provide $L_m = 1622$ uH in primary side**

RCD snubber

For details of formulas, please check the document [\[3\]](#).

Refer to the schematic above to see the snubber position.

$$L_{m_{uH}} = 1622$$

$$\text{Leakage} = 3.000 \times 10^{-2} \text{ (Hypothesis)}$$

$$\begin{aligned}
 L_{lk1_{uH}} &= L_{m_{uH}} \cdot \text{Leakage} \\
 &= 1622 \cdot 3.000 \times 10^{-2} \\
 &= 4.866 \times 10^1 \text{ (uH)}
 \end{aligned}$$

$$\begin{aligned}
 L_m &= L_{m_{uH}} \cdot 1 \times 10^{-6} \\
 &= 1622 \cdot 1 \times 10^{-6} \\
 &= 1.622 \times 10^{-3}
 \end{aligned}$$

$$\begin{aligned}
 L_{lk1} &= L_{lk1_{uH}} \cdot 1 \times 10^{-6} \\
 &= 4.866 \times 10^1 \cdot 1 \times 10^{-6} \\
 &= 4.866 \times 10^{-5}
 \end{aligned}$$

$$N_p = 26$$

$$N_s = 4$$

$$V_{breakDown} = 700 \text{ (see Tny-264 datasheet)}$$

$$\begin{aligned}
 V_{maxRating} &= 0.8 \cdot V_{breakDown} \\
 &= 0.8 \cdot 700 \\
 &= 5.600 \times 10^2 \text{ (20 percent of margin)}
 \end{aligned}$$

The snubber allowed voltage is

$$V_{sn} = V_{maxRating} - V_{in_{max}} = 560.000 - 400 = 160.000 \text{ (V)}$$

Chose the snubber resistor Rsn

$$\begin{aligned}
 V_{refl} &= (V_{out_{nom}} + V_d) \cdot n \\
 &= (12 + 0.400) \cdot 3.250 \\
 &= 40.300 \text{ (V)}
 \end{aligned}$$

$$\begin{aligned}
 FacV &= \frac{V_{sn}}{V_{sn} - V_{refl}} \\
 &= \frac{160.000}{160.000 - 40.300} \\
 &= 1.337
 \end{aligned}$$

$$\begin{aligned}
 R_{sn} &= \frac{(V_{sn})^2}{0.5 \cdot L_{lk1} \cdot (I_{peak})^2 \cdot Freq \cdot FacV} \\
 &= \frac{(160.000)^2}{0.5 \cdot 0.000 \cdot (0.250)^2 \cdot 132000.000 \cdot 1.337} \\
 &= 95415.312 \text{ (}\Omega\text{)}
 \end{aligned}$$

Let's chose Rsn = 82 kΩ

with this value the Vsn will be

$$R_{sn} = 82000.000 \text{ } (\Omega)$$

$$\begin{aligned} V_{sn} &= \frac{V_{refl} + \sqrt{(V_{refl})^2 + 2 \cdot L_{lk1} \cdot (I_{peak})^2 \cdot Freq \cdot R_{sn}}}{2} \\ &= \frac{40.300 + \sqrt{(40.300)^2 + 2 \cdot 0.000 \cdot (0.250)^2 \cdot 132000.000 \cdot 82000.000}}{2} \\ &= 150.016 \text{ (V)} \end{aligned}$$

The power of R_{sn} is

$$P_{R_{sn}} = \frac{(V_{sn})^2}{R_{sn}} = \frac{(150.016)^2}{82000.000} = 0.274 \text{ (W)}$$

Let's choose **$R_{sn} = 82k\Omega / 2W$** to avoid any under-designing.

- To define the snubber capacitor, let's start with allowed ripple of **5Vpp**

$$F_{min} = (132 - 8) \cdot 1 \times 10^3 = 124000.000 \text{ (kHz)}$$

$$\Delta V_{sn} = 5$$

$$C_{sn_{nF}} = 1 \times 10^9 \cdot \frac{V_{sn}}{\Delta V_{sn} \cdot R_{sn} \cdot F_{min}} = 1 \times 10^9 \cdot \frac{150.016}{5 \cdot 82000.000 \cdot 124000.000} = 2.951 \text{ (nF)}$$

$$C_{sn} = 1 \times 10^{-9} \cdot C_{sn_{nF}} = 1 \times 10^{-9} \cdot 2.951 = 0.000 \text{ (F)}$$

$$\Delta V_{sn_{percent}} = 100 \cdot \frac{\Delta V_{sn}}{V_{sn}} = 100 \cdot \frac{5}{150.016} = 3.333 \text{ (percent)}$$

Let's choose a **$C_{sn} = 10nF$**

With this value the ripple will be:

$$C_{sn} = 1.000 \times 10^{-8} \text{ (F)}$$

$$\Delta V_{sn} = \frac{V_{sn}}{C_{sn} \cdot R_{sn} \cdot F_{min}} = \frac{150.016}{0.000 \cdot 82000.000 \cdot 124000.000} = 1.475 \text{ (V)}$$

$$\Delta V_{sn_{percent}} = 100 \cdot \frac{\Delta V_{sn}}{V_{sn}} = 100 \cdot \frac{1.475}{150.016} = 0.983 \text{ (percent)}$$

The current of this capacitor will be

$$Z_{csn} = \frac{1}{2 \cdot \pi \cdot C_{sn} \cdot \text{Freq}} = \frac{1}{2 \cdot 3.142 \cdot 0.000 \cdot 132000.000} = 120.572 \text{ } (\Omega)$$

$$I_{Csn} = \frac{\Delta V_{sn}}{Z_{csn}} = \frac{1.475}{120.572} = 0.012 \text{ (App)}$$

$$I_{CsnRms} = \frac{I_{Csn}}{2 \cdot \sqrt{2}} = \frac{0.012}{2 \cdot \sqrt{2}} = 0.004 \text{ (Arms)}$$

$$I_{CsnRms_{mA}} = I_{CsnRms} \cdot 1 \times 10^3 = 0.004 \cdot 1 \times 10^3 = 4.326 \text{ (mArms)}$$

So the **Csn = 10nF** with minimum current of **5mA(rms)**

The output capacitor

Hypothesis of a sinusoidal current in the capacitor.

This hypothesis is not valid in our case, because the secondary current waveform in the transformer is a triangle that starts at its maximum value and decreases to zero, with no current during the charging phase. However, we will calculate the capacitor using this hypothesis just for information and comparison.

$$\text{Freq} = 132000.000 \text{ (Hz)} \quad V_{out_{nom}} = 12 \text{ (v)} \quad \text{Power}_{out} = 3.400 \text{ (w)}$$

$$\delta V_{out} = 0.010 \text{ (1 \%)} \quad D = 0.141 \text{ (duty cycle)}$$

Calculate the output capacitor for a target of 1% output voltage ripple.

$$I_{out} = \frac{\text{Power}_{out}}{V_{out_{nom}}} = \frac{3.400}{12} = 0.283 \text{ (A)}$$

$$\Delta V_{out} = \delta V_{out} \cdot V_{out_{nom}} = 0.010 \cdot 12 = 0.120 \text{ (V)}$$

$$C_{out_{uF}} = 1 \times 10^6 \cdot \frac{I_{out}}{\Delta V_{out} \cdot \text{Freq}} = 1 \times 10^6 \cdot \frac{0.283}{0.120 \cdot 132000.000} = 17.887 \text{ (uF)}$$

$$C_{out} = 1.789 \times 10^{-5} \text{ (F)}$$

$$Z_{Cout} = \frac{1}{C_{out} \cdot 2 \cdot \pi \cdot \text{Freq}} = \frac{1}{0.000 \cdot 2 \cdot 3.142 \cdot 132000.000} = 0.067 \text{ } (\Omega)$$

$$I_{Cout} = \frac{\Delta V_{out}}{Z_{Cout}} = \frac{0.120}{0.067} = 1.780 \text{ (App)}$$

$$I_{Cout_{RMS}} = \frac{I_{Cout}}{2 \cdot \sqrt{2}} = \frac{1.780}{2 \cdot \sqrt{2}} = 0.629 \text{ (Arms)}$$

Output capacitor of the flyback using the real waveforms

Input data for the calculation

$$L_m = 1.622 \times 10^{-3} \text{ (H)} \quad N_p = 26 \quad N_s = 4$$

$$\delta_{V_{out}} = 1.000 \times 10^{-2} \text{ (1 \%)} \quad V_{out_{nom}} = 12 \text{ (v)} \quad V_d = 4.000 \times 10^{-1} \text{ (Diode Forward Drop in V)}$$

$$\Delta_{V_{out}} = \delta_{V_{out}} \cdot V_{out_{nom}} = 0.010 \cdot 12 = 0.120 \text{ (V)}$$

$$n = \frac{N_p}{N_s} = \frac{26}{4} = 6.500$$

$$V_o = V_{out_{nom}} + V_d = 12 + 0.400 = 12.400 \text{ (V)}$$

$$I_{peak} = 0.250 \text{ (A, Primary fixed by the controller limit)}$$

Calculate the secondary inductance and ON time

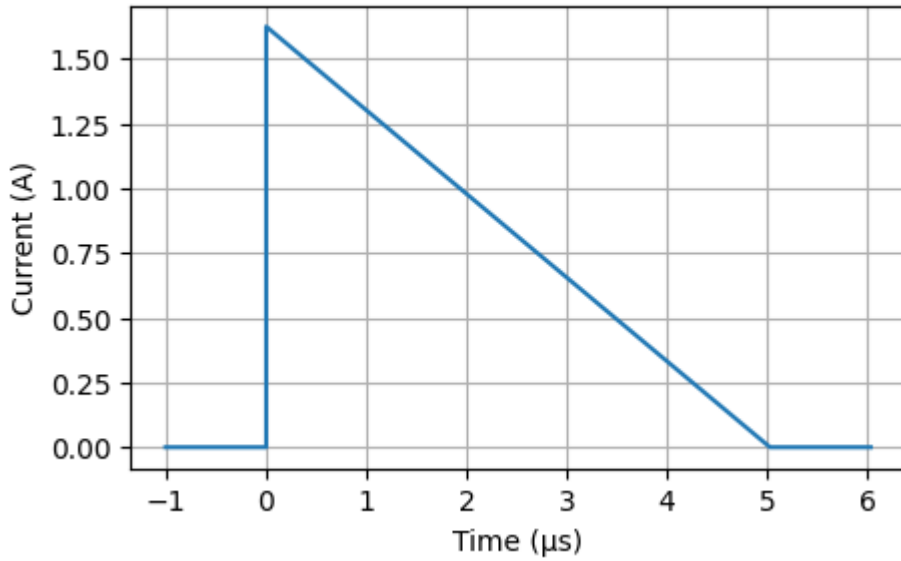
$$\begin{aligned} L_{ms} &= \frac{L_m}{(n)^2} \\ &= \frac{1.622 \times 10^{-3}}{(6.500 \times 10^0)^2} \\ &= 3.839 \times 10^{-5} \text{ (secondary magnetic inductance, H)} \end{aligned}$$

$$\begin{aligned} I_{s_{peak}} &= n \cdot I_{peak} \\ &= 6.500 \times 10^0 \cdot 2.500 \times 10^{-1} \\ &= 1.625 \times 10^0 \text{ (A , secondary)} \end{aligned}$$

$$\begin{aligned} dI_{sec} &= I_{s_{peak}} - 0 \\ &= 1.625 \times 10^0 - 0 \\ &= 1.625 \times 10^0 \end{aligned}$$

$$\begin{aligned} dt_{Sec} &= L_{ms} \cdot \frac{dI_{sec}}{V_o} \\ &= 3.839 \times 10^{-5} \cdot \frac{1.625 \times 10^0}{1.240 \times 10^1} \\ &= 5.031 \times 10^{-6} \text{ (in seconds , reminder } v=Ldi/dt) \end{aligned}$$

Secondary current of the flyback (IDEAL waveform)



Current in the capacitor

$$i = C \frac{dv_{out}}{dt}$$

The secondary current decreases linearly from I_0 to 0 during dt_{sec} , where $I_0 = n \cdot I_{peak}$ is the secondary peak current:

$$i(t) = I_0 \left(1 - \frac{t}{dt_{sec}} \right)$$

Voltage variation on the capacitor:

$$\Delta V_{out} = \frac{1}{C} \int_0^{dt_{sec}} i(t) dt$$

$$\Delta V_{out} = \frac{1}{C} \int_0^{dt_{sec}} I_0 \left(1 - \frac{t}{dt_{sec}} \right) dt$$

$$\Delta V_{out} = \frac{1}{C} \left[I_0 t - \frac{I_0}{2 dt_{sec}} t^2 \right]_0^{dt_{sec}}$$

$$\Delta V_{out} = \frac{1}{C} \left(I_0 dt_{sec} - \frac{I_0}{2} dt_{sec} \right)$$

$$\Delta V_{out} = \frac{I_0 dt_{sec}}{2C}$$

$$\text{Using } dt_{sec} = \frac{L_{ms} \Delta I_{sec}}{V_o}$$

where $V_o = V_{out} + V_d$ is the voltage across the transformer secondary (including the diode drop), and $\Delta I_{sec} = I_0$:

$$\Delta V_{out} = \frac{I_0^2 L_{ms}}{2C V_o}$$

Since $L_{ms} = \frac{L_m}{n^2}$ and $I_0 = n \cdot I_{peak}$:

$$\Delta V_{out} = \frac{n^2 I_{peak}^2 \cdot \frac{L_m}{n^2}}{2C V_o} = \frac{L_m I_{peak}^2}{2C V_o}$$

And

$$C = \frac{L_m I_{peak}^2}{2 V_o \Delta V_{out}}$$

where $V_o = V_{out} + V_d$

$$\begin{aligned} C_{out} &= L_m \cdot \frac{(I_{peak})^2}{2 \cdot V_o \cdot \Delta V_{out}} \\ &= 1.622 \times 10^{-3} \cdot \frac{(2.500 \times 10^{-1})^2}{2 \cdot 1.240 \times 10^1 \cdot 1.200 \times 10^{-1}} \\ &= 3.406 \times 10^{-5} \end{aligned}$$

$$C_{ouF} = C_{out} \cdot 1 \times 10^6 = 0.000 \cdot 1 \times 10^6 = 34.064 \text{ (uF)}$$

For the current in the capacitor, we assume that the output current has no AC components, so all the AC ripple flows into the capacitor. Therefore, the capacitor's peak current is equal to the peak current delivered by the secondary winding.

$$I_{c_{peak}} = 1.625 \text{ (A)}$$

Using the RMS definition to compute the $I_{s_{rms}}$.

$$I_{s_{rms}} = \sqrt{\frac{1}{T} \int_0^T I_s(t)^2 dt}$$

$$I_{s_{rms}} = \sqrt{\frac{1}{T} \int_0^{dt_{sec}} I_{s_{peak}}^2 \left(1 - \frac{t}{dt_{sec}}\right)^2 dt} = I_{s_{peak}} \sqrt{\frac{dt_{sec}}{3T}}$$

So in our case

$$T_{period} = \frac{1}{\text{Freq}} = \frac{1}{1.320 \times 10^5} = 7.576 \times 10^{-6} \text{ (s)}$$

$$I_{s_{rms}} = I_{s_{peak}} \cdot \sqrt{\frac{dt_{sec}}{3 \cdot T_{period}}} = 1.625 \times 10^0 \cdot \sqrt{\frac{5.031 \times 10^{-6}}{3 \cdot 7.576 \times 10^{-6}}} = 7.646 \times 10^{-1} \text{ (A)}$$

Below is a comparison between the simplified sinusoidal assumption and the more representative waveform of the flyback secondary.

	Sinusoidal	Flyback Waveform
Min Co (μF)	17.890	34.0640
Capa Peak-Peak Current (App)	1.780	1.6250
Capa RMS current (Arms)	0.629	0.7646

Verification:

$$\Delta V_{out} = L_m \cdot \frac{(I_{peak})^2}{2 \cdot C_{out} \cdot V_o} = 0.002 \cdot \frac{(0.250)^2}{2 \cdot 0.000 \cdot 12.400} = 0.120$$

$$\delta_{V_{out}} = 100 \cdot \frac{\Delta V_{out}}{V_{out_{nom}}} = 100 \cdot \frac{0.120}{12} = 1.000 (\%)$$

Same value $\delta_{V_{out}}$ of our target

The selected capacitor

We will use the capaitor **EEVFK1V471Q 470uF 35V 20%**

For more information pease check the datasheet below [1].

FK series

Characteristics list

Endurance : 105 °C 2000 h (ø12.5 : 5000 h)

Rated voltage (V)	Capacitance (±20 %) (μF)	Case size (mm)			Size code ^{*1}	Specification			Part No.		Reflow	Min. Packaging Q'ty (pcs)
		øD	L			Ripple current ^{*2} (mA rms)	Impe- dance ^{*3} (Ω)	tan δ ^{*4}	Standard Product	Vibration-proof product		Taping
			Standard	Vibration- proof								
35	4.7	4.0	5.8	—	B	90	1.35	0.12	EEEFK1V4R7R	—	(1)	2000
	10	4.0	5.8	—	(B)	90	1.35	0.12	EEEFK1V100UR	—	(1)	2000
		5.0	5.8	—	C	160	0.70	0.12	EEEFK1V100R	—	(1)	1000
	22	5.0	5.8	—	C	160	0.70	0.12	EEEFK1V220R	—	(1)	1000
	33	6.3	5.8	6.1	D	240	0.36	0.12	EEEFK1V330P	EEEFK1V330V	(1)	1000
	47	6.3	5.8	6.1	D	240	0.36	0.12	EEEFK1V470P	EEEFK1V470V	(1)	1000
	68	6.3	7.7	8	D8	280	0.34	0.12	EEEFK1V680XP	EEEFK1V680XV	(1)	900
	100	6.3	7.7	8	D8	280	0.34	0.12	EEEFK1V101XP	EEEFK1V101XV	(1)	900
		8.0	10.2	10.5	F	600	0.16	0.12	EEEFK1V101P	EEEFK1V101V	(2)	500
	150	8.0	10.2	10.5	F	600	0.16	0.12	EEEFK1V151P	EEEFK1V151V	(2)	500
	220	8.0	10.2	10.5	F	600	0.16	0.12	EEEFK1V221P	EEEFK1V221V	(2)	500
	330	10.0	10.2	10.5	G	850	0.08	0.12	EEEFK1V331P	EEEFK1V331V	(2)	500
	470	12.5	13.5	13.8	H13	1100	0.06	0.12	EEVFK1V471Q	EEVFK1V471V	(3)	200
	680	12.5	13.5	13.8	H13	1100	0.06	0.12	EEVFK1V681Q	EEVFK1V681V	(3)	200

*1: Size code(): Miniaturization product

*2: Ripple current (100 kHz / +105 °C)

*3: Impedance (100 kHz / +20 °C)

*4: tan δ (120 Hz / +20 °C)

• Please refer to the page of "Reflow Profile" and "The Taping Dimensions".

Figure 7: Screenshot of the output capacitor datasheet

ESR estimation from $\tan(\delta)$

The dissipation factor (loss tangent) of a capacitor is defined as:

$$\tan(\delta) = \frac{ESR}{X_C} = ESR \cdot \omega \cdot C = ESR \cdot 2\pi f \cdot C$$

So:

$$ESR = \frac{\tan(\delta)}{2\pi f \cdot C}$$

$$C_{nom_{esr}} = 4.700 \times 10^{-4} \text{ (F, nominal capacitance)}$$

$$tol = 2.000 \times 10^{-1} \text{ (20\% tolerance)}$$

$$\tan_{\delta} = 1.200 \times 10^{-1} \text{ (dissipation factor at 120 Hz, +20°C)}$$

$$f_{tan} = 120 \text{ (Hz, frequency of tan(delta) spec)}$$

$$C_{min} = C_{nom_{esr}} \cdot (1 - tol) = 4.700 \times 10^{-4} \cdot (1 - 2.000 \times 10^{-1}) = 3.760 \times 10^{-4} \text{ (F)}$$

$$ESR_{max} = \frac{\tan_{\delta}}{2 \cdot \pi \cdot f_{tan} \cdot C_{min}} = \frac{1.200 \times 10^{-1}}{2 \cdot 3.142 \times 10^0 \cdot 120 \cdot 3.760 \times 10^{-4}} = 4.233 \times 10^{-1} \text{ (Ohm)}$$

Since we don't have ESR at heigh frequency, we will use the simple ESR formula below:

$$\text{ESR frequency scaling for electrolytics [5]: } ESR(f) = ESR(120Hz) \cdot \sqrt{\frac{120Hz}{f}}$$

$$ESR_{switching} = ESR_{max} \cdot \sqrt{\frac{120}{Fmin}} = 4.233 \times 10^{-1} \cdot \sqrt{\frac{120}{1.240 \times 10^5}} = 1.317 \times 10^{-2} \text{ (ohm)}$$

The power dissipation at the capacitor:

$$\begin{aligned} P_{Co} &= (I_{rms})^2 \cdot ESR_{switching} \\ &= (7.646 \times 10^{-1})^2 \cdot 1.317 \times 10^{-2} \\ &= 7.697 \times 10^{-3} \text{ (W)} \end{aligned}$$

We don't have the thermal resistance of this capacitor, nor the maximum temperature associated with the maximum current. In the datasheet, we can see that the capacitor's Tmax is 105°C, which corresponds to its maximum current.

Let's assume that the temperature at the maximum current is 100°C.

So can estimate the R_{th} as:

$$I_{rmsMax}_{datasheet} = 1.100 \text{ (Arms from the datasheet @ 100 kHz / +105 °C)}$$

$$T_{max} = 105 \text{ (°C from the datasheet)}$$

$$T_{maxI} = 100 \text{ (°C, Hypothesis, max temperature at } I_{rmsMax})$$

$$\begin{aligned} P_{maxI} &= (I_{rmsMax}_{datasheet})^2 \cdot ESR_{switching} \\ &= (1.100)^2 \cdot 0.013 \\ &= 0.016 \text{ (W)} \end{aligned}$$

$$\begin{aligned} R_{th} &= \frac{T_{max} - T_{maxI}}{P_{maxI}} \\ &= \frac{105 - 100}{0.016} \\ &= 313.814 \text{ (°C/W)} \end{aligned}$$

For our operating point, the capacitor's T_{max} can be estimated using this relation.

$$T_{maxCo} = P_{Co} \cdot R_{th} + T_{maxI} = 0.008 \cdot 313.814 + 100 = 102.415 \text{ (°C)}$$

Let's estimate the ΔV_{out} in the worst case. We will assume that ΔV_{out} is simply the sum of the maximum voltage caused by the capacitor and the maximum voltage caused by the ESR.

$$\begin{aligned} \Delta V_{outCAPA} &= Lm \cdot \frac{(I_{peak})^2}{2 \cdot C_{min} \cdot V_o} \\ &= 1.622 \times 10^{-3} \cdot \frac{(2.500 \times 10^{-1})^2}{2 \cdot 3.760 \times 10^{-4} \cdot 1.240 \times 10^1} \\ &= 1.087 \times 10^{-2} \text{ (V, Pure capacitor)} \end{aligned}$$

$$\begin{aligned} \Delta V_{outESR} &= I_{speak} \cdot ESR_{switching} \\ &= 1.625 \times 10^0 \cdot 1.317 \times 10^{-2} \\ &= 2.140 \times 10^{-2} \text{ (V, Pure resistance)} \end{aligned}$$

$$\begin{aligned} \Delta V_{outWORST} &= \Delta V_{outCAPA} + \Delta V_{outESR} \\ &= 1.087 \times 10^{-2} + 2.140 \times 10^{-2} \\ &= 3.227 \times 10^{-2} \text{ (V)} \end{aligned}$$

Above is a simplified calculation but it is a maximum of what ΔV_{out} can be.

Let's check the relative voltage delta:

$$\delta V_{outWORST} = 0.269 \text{ (\% < 1\% OK)}$$

The final flyback Schematic & Modification

Below, you will find the original schematic followed by the updated version after the modifications.

Schematic

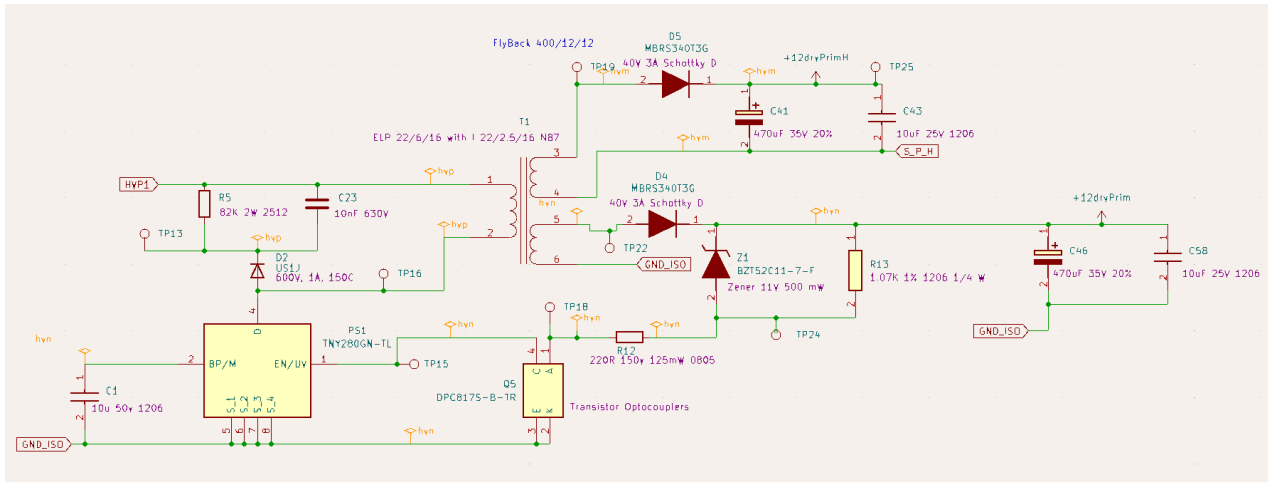


Figure 8: The initial schematic of the flyback: 2 outputs to power the primary driver

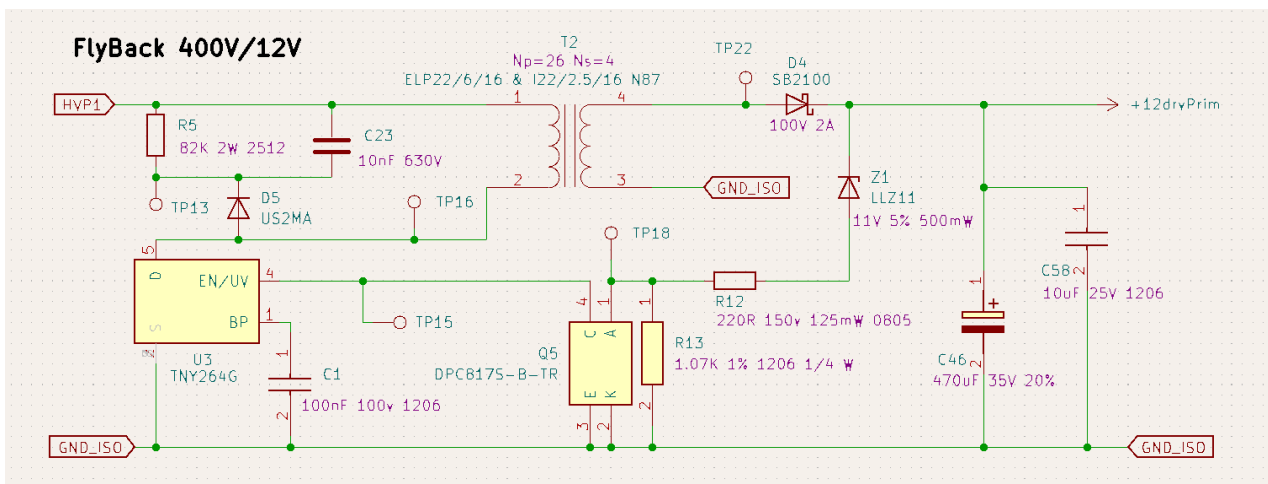


Figure 9: The final flyback schematic with one output

Modifications

Below a table that resume all modifications in flyback circuit:

Change	Before	After	Why	Impact
IC controller	TNY280	TNY264	I_{lim} in TNY280 = 750 mA vs 250 mA for TNY264 Risk of core saturation	Not < pin-to-pin equivalent PCB modification
Output diode	MBRS340T3G	SB2100	VRRM = 40 V (old) vs 100 V (new) VRRM of new design ≈ 62 V (peak)	Not SMD PCB adaptation

Change	Before	After	Why	Impact
Snubber diode	U51J	US2MA	VRRM = 600 V (old) vs 1 kV (new) More margin on VRRM	No PCB impact
Transformer air gap	No air gap	Add air gap	Avoid core saturation	Low cycle energy (low power)
Transformer ratio	Np:Ns1:Ns2 = 8:1:1	Np:Ns = 26:4	Avoid core saturation	PCB impact External winding needed Primary winding: 8 turns on PCB + 18 turns with wire.
R13 (optocoupler parallel resistor)	Parallel with zener	Parallel with opto LED	Mistake in initial schematic	PCB impact
Zener	BZT52C11	LLZ11	Out of stock	Pin-to-pin compatible
BP capacitor C1	10 μ F	100 nF	Result of TNY280 $\rightarrow$ TNY264	Pin-to-pin compatible

PCB flyback zone

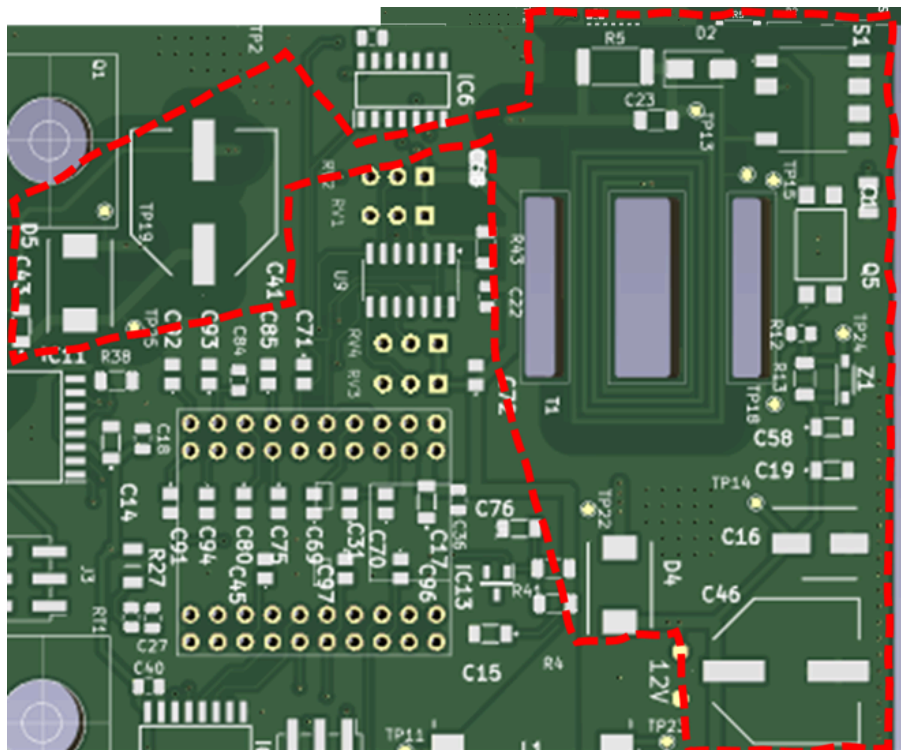


Figure 10: Naked PCB: The PCB flyback zone

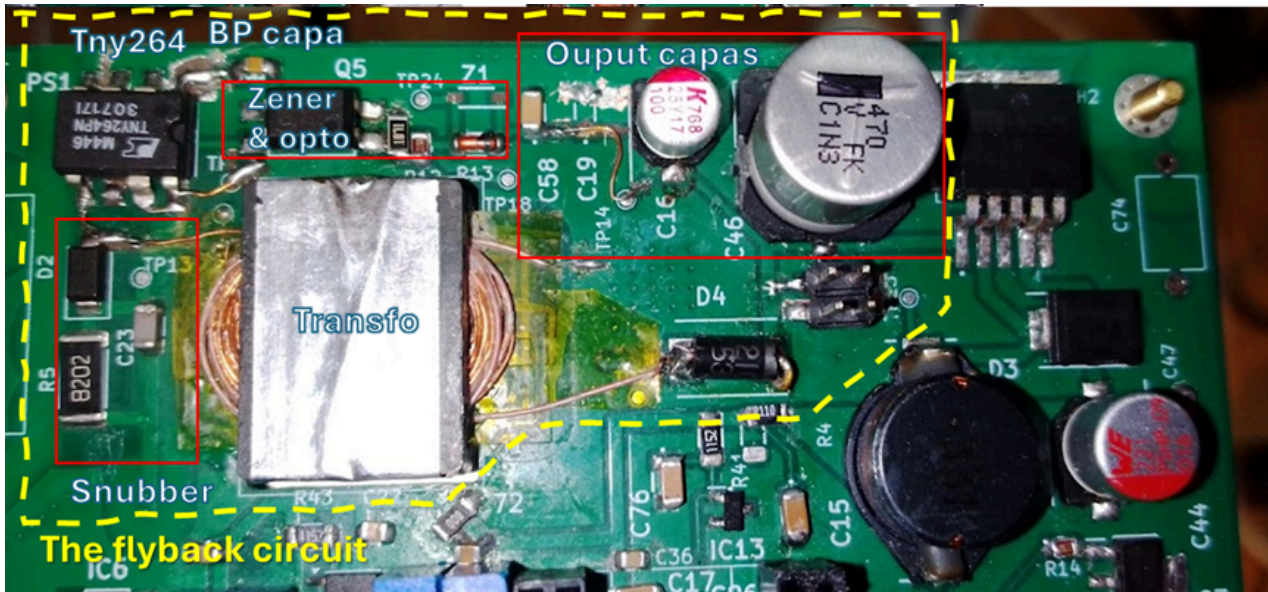


Figure 11: The flyback with all modifications

As mentioned in the table above, the initial flyback is modified, so the final version is presented in the figure above (fast prototyping with manual modification of the PCB)

Test & Measurements

Setup

Below is the setup schematic displaying all the signal names used in this part.

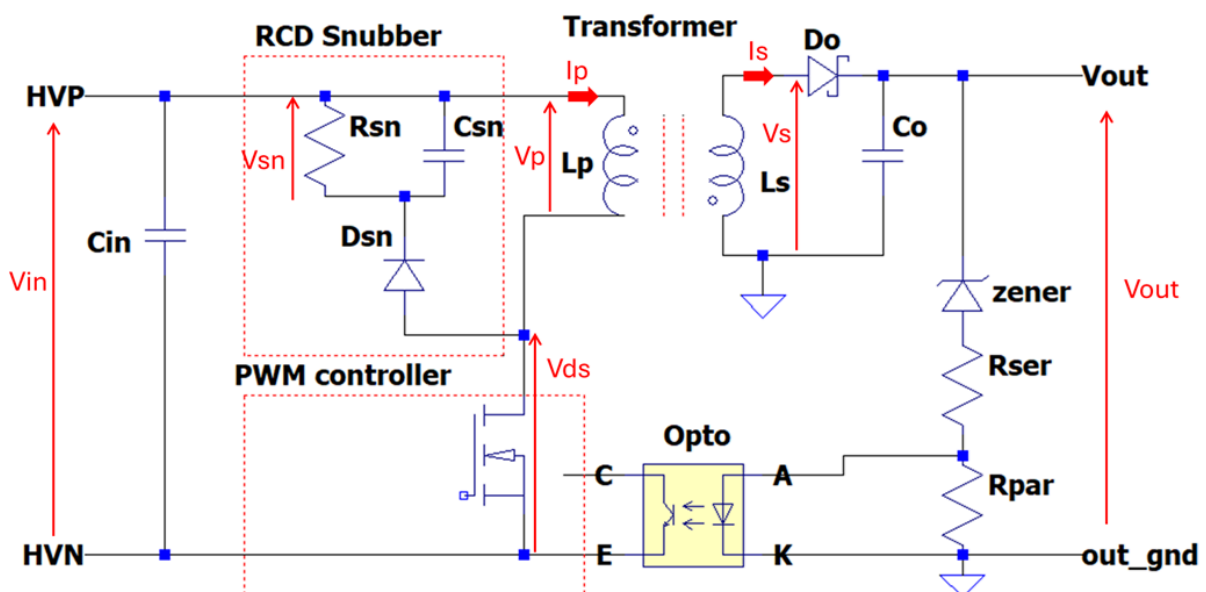


Figure 12: TEST

Small-signal transformer characterization

In this part, we will characterize the transformer core using small signal only

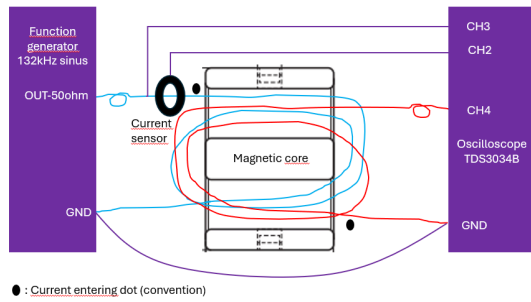


Figure 13: Test Setup

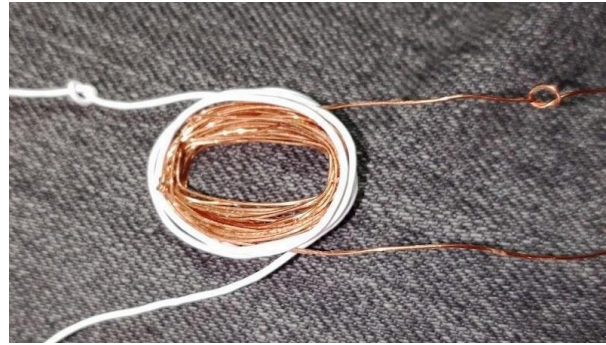


Figure 14: The transformer winding

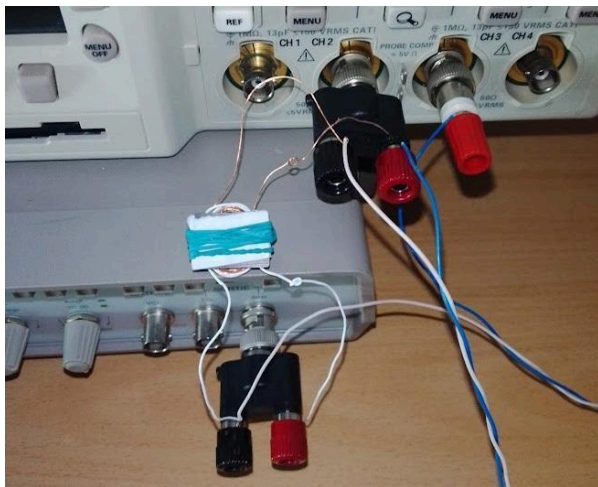


Figure 15: Photo of the setup

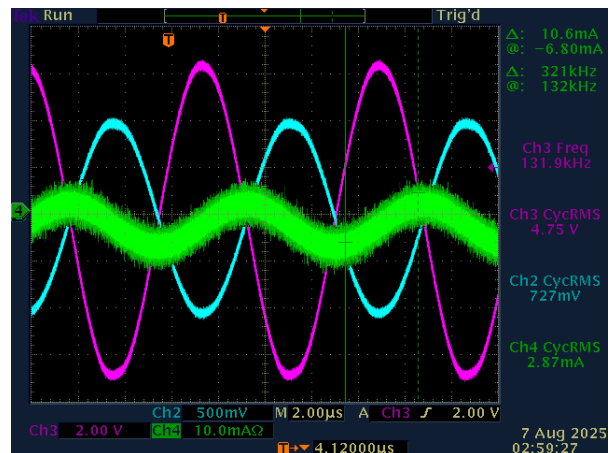


Figure 16: Oscilloscope ScreenShot

Result analysis

In this section, we will analyze the measured signals (V_p , V_s , I_p) as well as the estimated signals (V_s -estimation and I_p -estimation):

- The V_s -estimation allows us to validate the transformer turns ratio N_p/N_s
- The I_p -estimation enables us to assess the inductance in small-signal conditions

ELP 22/6/16 with I 22/2.5/16 N87: Small-signal measurement Np=26, Ns=4, no air gap

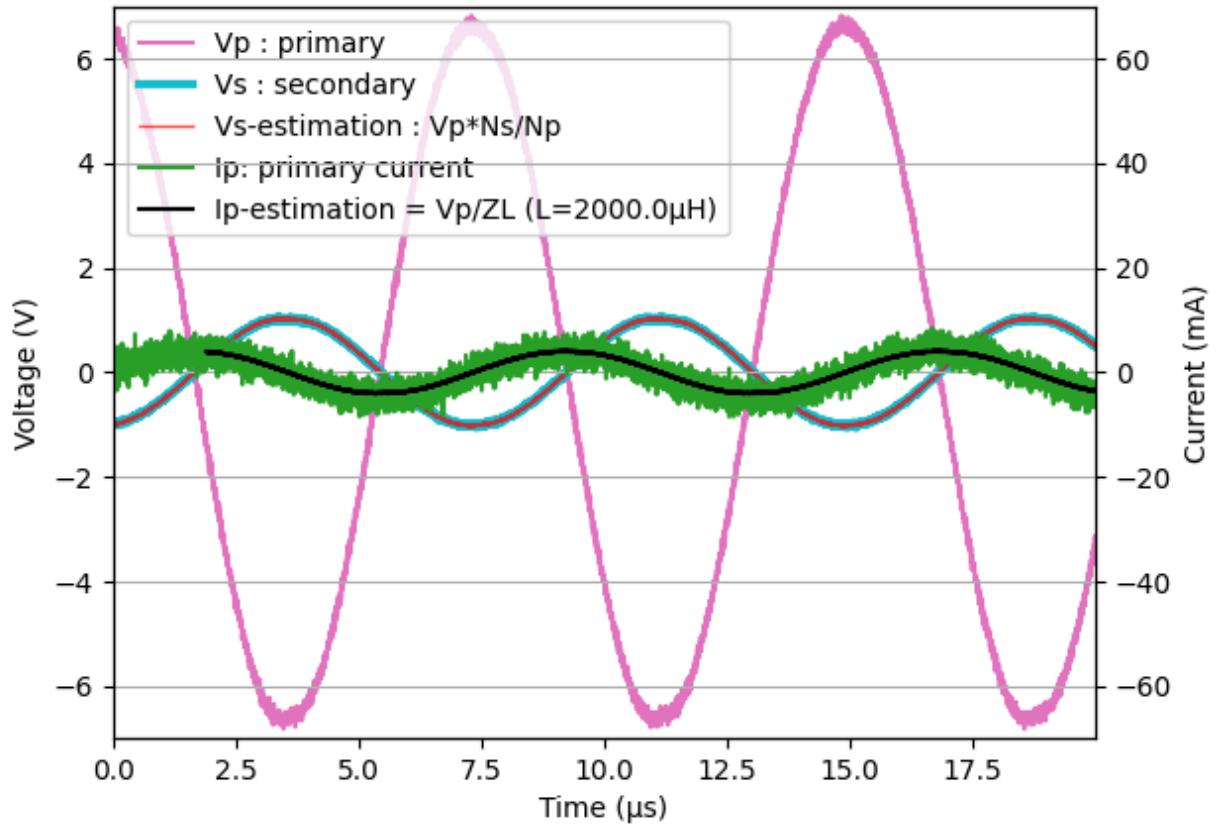


Figure 17: TEST

The results show that:

- The estimated Vs aligns closely with the measured Vs (blue vs red), confirming that the turn ratio (Np, Ns) = (26, 4) is valid
- The estimated Ip aligns closely the measured Ip (green vs black), suggesting that the primary inductance Lp is around 2000 μH

Below a simulation of this test using LT-Spice

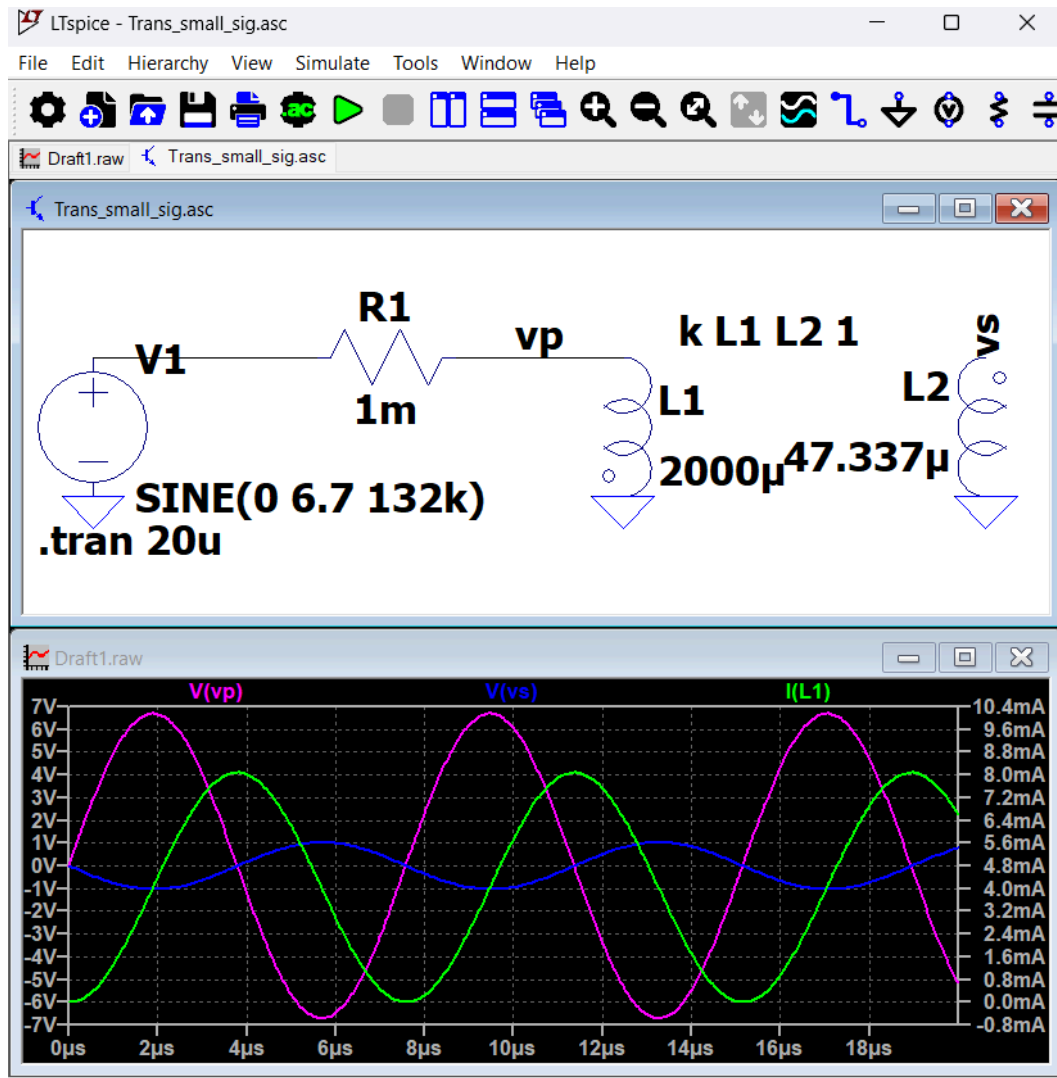


Figure 18: TEST

Below the calculation of the AL (nH/turns²)

$$N_p = 26$$

$$L_{p_{uH}} = 2000 \text{ (uH)}$$

$$\begin{aligned} L_{p_{nH}} &= L_{p_{uH}} \cdot 1 \times 10^3 \\ &= 2000 \cdot 1 \times 10^3 \\ &= 2000000.000 \text{ (nH)} \end{aligned}$$

$$\begin{aligned} AL_{nH} &= \frac{L_{p_{nH}}}{(N_p)^2} \\ &= \frac{2000000.000}{(26)^2} \\ &= 2958.580 \text{ (nH/turns}^2\text{)} \end{aligned}$$

$$AL_{nom} = 5200 \text{ (nH/turns}^2\text{ from the datasheet)}$$

$$\begin{aligned} \text{diff} &= 100 \cdot \frac{AL_{nH} - AL_{nom}}{AL_{nom}} \\ &= 100 \cdot \frac{2958.580 - 5200}{5200} \\ &= -43.104 \text{ (\% > 25\% of the datasheet tolerance)} \end{aligned}$$

The difference between the measured values and the datasheet specifications can be attributed to the air gap present in our setup.

To quantify this, we will estimate the equivalent air gap, assuming that the core's AL value matches the datasheet specification [1].

$$K1 = 134 \text{ (see the core datasheet)}$$

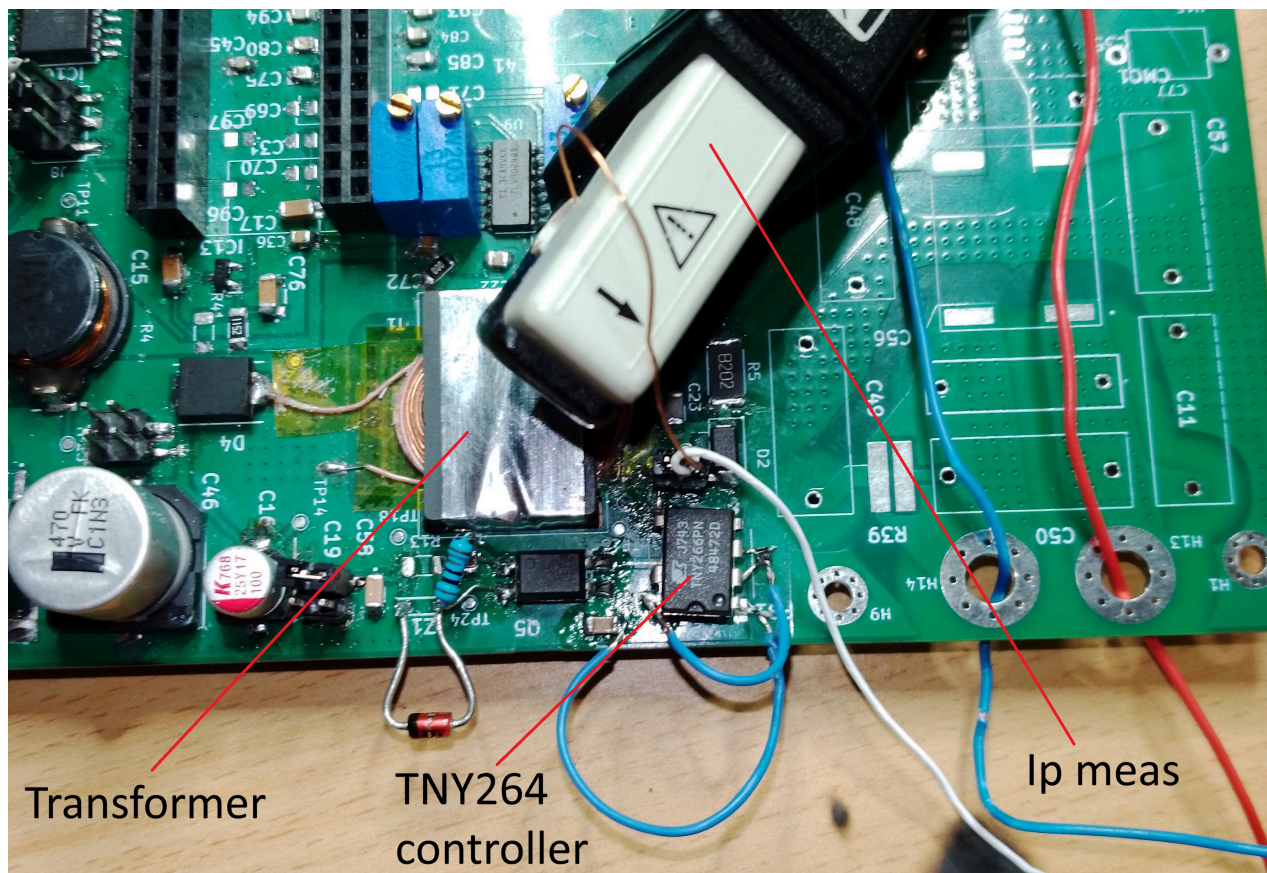
$$K2 = -0.806 \text{ (see the core datasheet)}$$

$$\text{air}_{gap} = \left(\frac{AL_{nH}}{K1} \right)^{\left(\frac{1}{K2} \right)} = \left(\frac{2958.580}{134} \right)^{\left(\frac{1}{-0.806} \right)} = 0.022 \text{ (mm)}$$

Therefore, an effective air gap of 22 μm is reasonable, implying a physical gap of approximately 11 μm , as the magnetic flux passes through it twice, producing the 22 μm equivalent.

Primary current and voltage

Setup



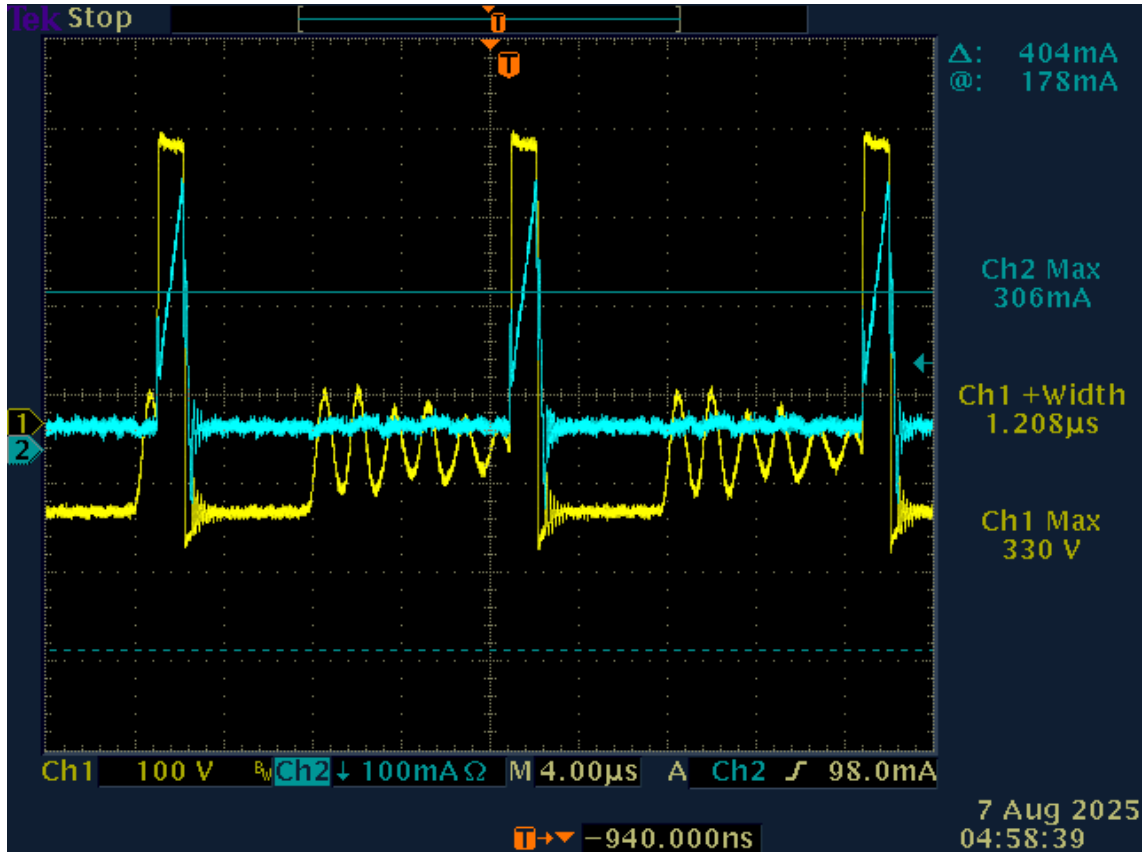


Figure 20: The measurement of I_p and V_p (oscilloscope screenshot)

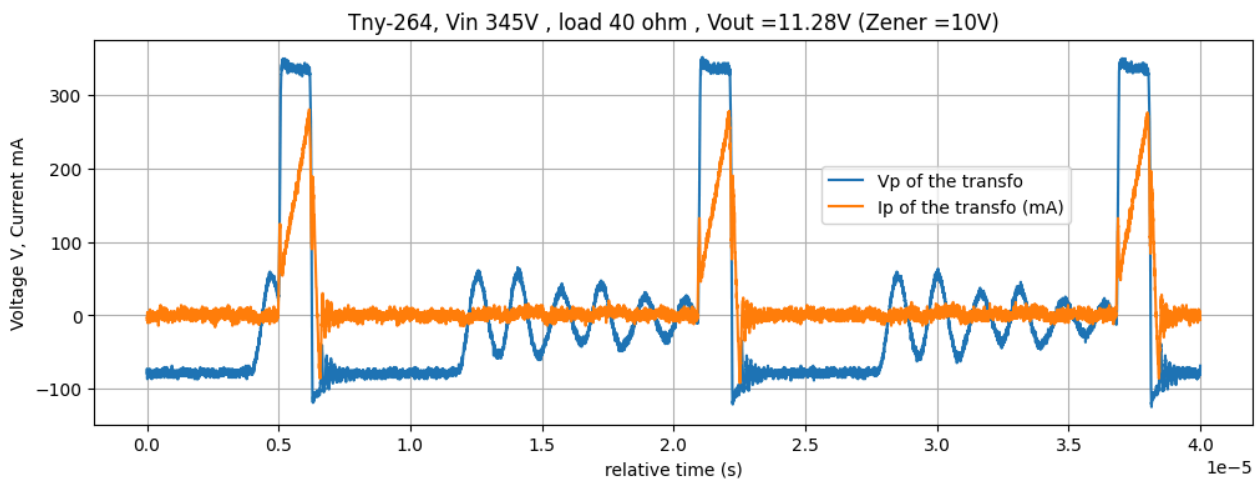


Figure 21: The measurement of I_p and V_p (oscilloscope data)

Transformer characterization

In this preprocessing step, we isolate the ON phase of the primary MOSFET and estimate L_m and R_{ds} :

- L_m , the primary magnetizing inductance, is estimated from the slope of the current during the ON phase ($v = L di/dt$), as shown by the blue and black curves
- R_{ds} , the MOSFET's on-resistance, is estimated from the voltage drop at the ON phase ($v = R_{ds} \times I$) using linear regression to reduce noise, as shown by the red and green curves.

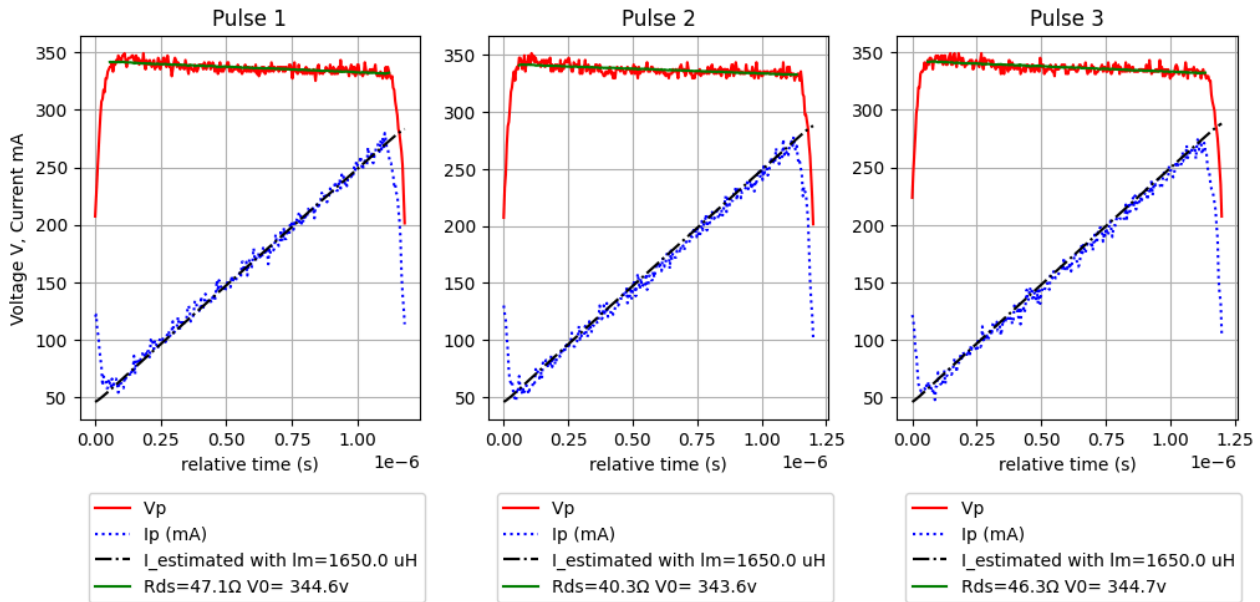


Figure 22: Pre-Processing of the measurement: we extract the L_m and R_{ds_on} from the I_p and V_p waveforms

From the above figure, the estimated L_m and R_{ds} are:

The average R_{ds_on} value is 44.5Ω (see below), which falls within the datasheet specification range of $[28 \Omega, 48 \Omega]$, see[1].

The average V_{in} is $344.3 V$, and the measured value is about $345 V$, which indicates good quality in the linear regression estimation.

$$L_m = 1650 \text{ (}\mu H\text{)}$$

$$R_{ds} = [47.100, 40.300, 46.300] \text{ (}\Omega\text{)}$$

$$R_{ds_{avg}} = 44.567 \text{ (}\Omega\text{)}$$

$$V_{in} = [344.600, 343.600, 344.700] \text{ (V)}$$

$$V_{in_{avg}} = 344.300 \text{ (V)}$$

Let's estimate the AL and the airgap.

$$L_{p_{nH}} = L_m \cdot 1 \times 10^3 = 1650 \cdot 1 \times 10^3 = 1650000.000 \text{ (nH)}$$

$$AL_{nH} = \frac{L_{p_{nH}}}{(N_p)^2} = \frac{1650000.000}{(26)^2} = 2440.828 \text{ (nH/turns}^2\text{)}$$

$$K1 = 134 \text{ (see the core datasheet)}$$

$$K2 = -0.806 \text{ (see the core datasheet)}$$

$$air_{gap} = \left(\frac{AL_{nH}}{K1} \right)^{\left(\frac{1}{K2} \right)} = \left(\frac{2440.828}{134} \right)^{\left(\frac{1}{-0.806} \right)} = 0.027 \text{ (mm)}$$

So the above estimation suggests an air gap of 27 μm

Estimation of the leakage inductance and the primary MOSFET capacitor

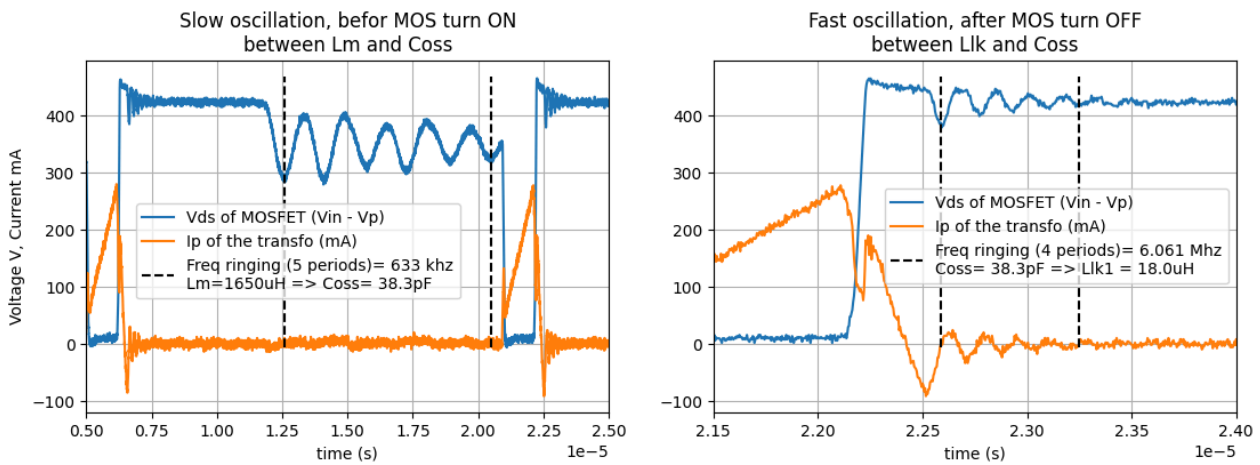


Figure 23: Zoom on the Vp signal, slow/fast oscillation of Vp

Coss estimation (left graph).

$$L_m = 1650 \text{ (uH see above estimation)}$$

$$t_0 = 1.260 \times 10^{-5} \text{ (see left graph above, black instant)}$$

$$t_1 = 2.050 \times 10^{-5} \text{ (see left graph above, black instant)}$$

$$n = 5 \text{ (Number of periods)}$$

$$fring = \frac{n}{t_1 - t_0} = \frac{5}{2.050 \times 10^{-5} - 1.260 \times 10^{-5}} = 6.329 \times 10^5 \text{ (Ring frequency in Hz)}$$

$$\begin{aligned} C_{oss} &= \frac{1}{(2 \cdot \pi)^2 \cdot L_m \cdot 1 \times 10^{-6} \cdot (fring)^2} \\ &= \frac{1}{(2 \cdot 3.142 \times 10^0)^2 \cdot 1650 \cdot 1 \times 10^{-6} \cdot (6.329 \times 10^5)^2} \\ &= 3.832 \times 10^{-11} \text{ (F)} \end{aligned}$$

Conventional units

$$\text{fring}_{MHz} = \text{fring} \cdot 1 \times 10^{-6} = 632911.392 \cdot 1 \times 10^{-6} = 0.633 \text{ (MHz)}$$

$$\text{Coss}_{pF} = \text{Coss} \cdot 1 \times 10^{12} = 0.000 \cdot 1 \times 10^{12} = 38.324 \text{ (pF)}$$

Primary leakage inductance estimation (right graph).

$$t_0 = 2.259 \times 10^{-5} \text{ (see right graph above, black instant)}$$

$$t_1 = 2.325 \times 10^{-5} \text{ (see right graph above, black instant)}$$

$$n = 4 \text{ (Number of periods)}$$

$$\text{fring} = \frac{n}{t_1 - t_0} = \frac{4}{2.325 \times 10^{-5} - 2.259 \times 10^{-5}} = 6.061 \times 10^6 \text{ (Ring frequency in Hz)}$$

$$\begin{aligned} \text{Llk1} &= \frac{1}{(2 \cdot \pi)^2 \cdot \text{Coss} \cdot (\text{fring})^2} \\ &= \frac{1}{(2 \cdot 3.142 \times 10^0)^2 \cdot 3.832 \times 10^{-11} \cdot (6.061 \times 10^6)^2} \\ &= 1.799 \times 10^{-5} \text{ (H)} \end{aligned}$$

Conventional units

$$\text{fring}_{MHz} = \text{fring} \cdot 1 \times 10^{-6} = 6060606.061 \cdot 1 \times 10^{-6} = 6.061 \text{ (MHz)}$$

$$\text{Llk1}_{uH} = \text{Llk1} \cdot 1 \times 10^6 = 0.000 \cdot 1 \times 10^6 = 17.994 \text{ (uH)}$$

$$\text{Prim}_{RelativeLeak} = 100 \cdot \frac{\text{Llk1}_{uH}}{\text{Lm}} = 100 \cdot \frac{17.994}{1650} = 1.091 \text{ (\% relative leakage inductance)}$$

So 1% of leakage in primary is normal value

Verification of the max duty cycle

For this test, we use the same setup as before, but we lower the input voltage to 48V to force the TNY-264 controller into its maximum duty-cycle operation.

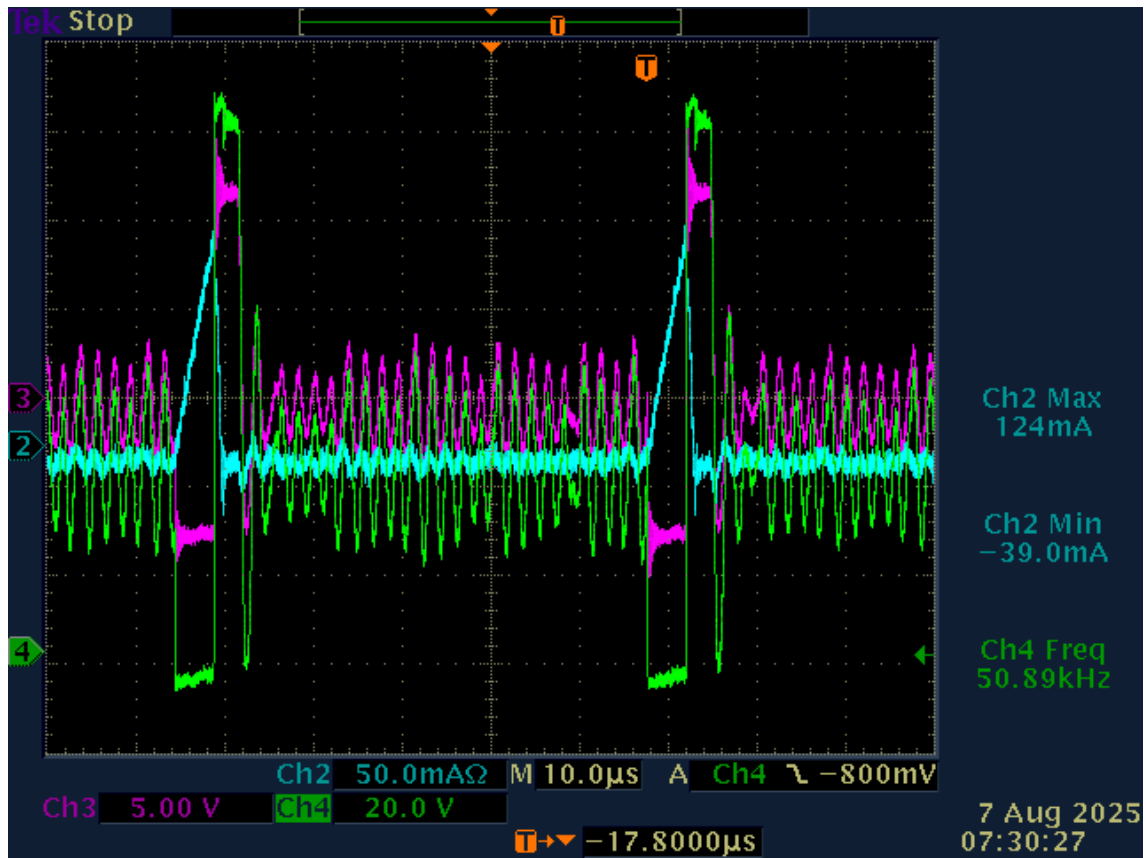


Figure 24: Oscilloscope screenshot

Tny-264 Vin = 47.50 V

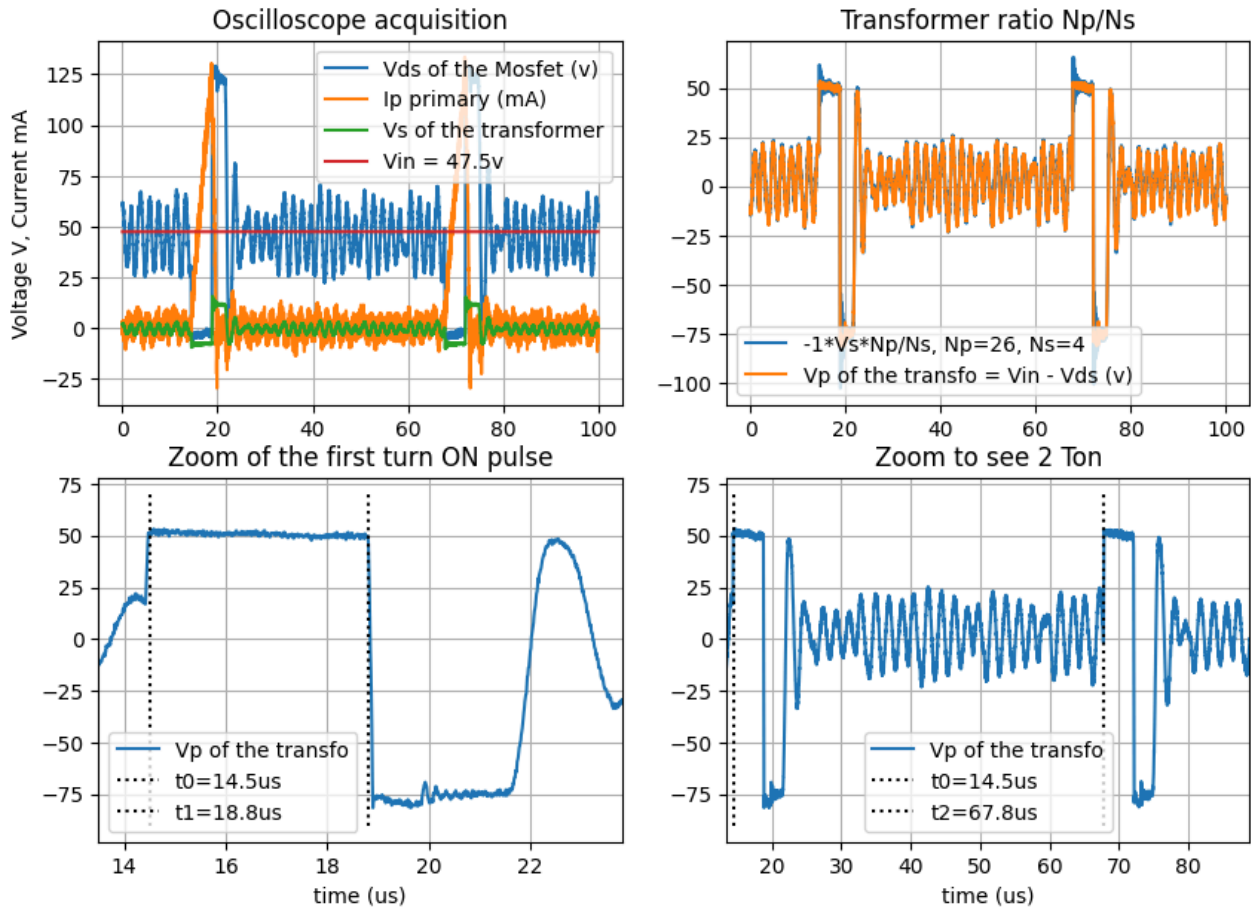


Figure 25: Oscilloscope data at 47.5V input: Pre-Processing of the measurement

From the preprocessing of the measurements:

- top-left graph: raw data, showing a primary peak current of roughly 125 mA, below the 250 mA limit—which confirms that the cycle is reset by reaching the maximum duty cycle
- top-right graph: confirmation of the N_p/N_s turns ratio
- bottom-left: zoom on the T_{on} interval, used to compute T_{on} from t_0 and t_1 (see calculation below)
- bottom-right: zoom showing two consecutive T_{on} intervals, used to estimate the switching frequency from t_0 and t_2 (see calculation below)

$$t_0 = 14.500 \text{ } (\mu\text{s})$$

$$t_1 = 18.800 \text{ } (\mu\text{s})$$

$$t_2 = 67.800 \text{ } (\mu\text{s})$$

$$T_{on} = t_1 - t_0 = 18.800 - 14.500 = 4.300 \text{ } (\mu\text{s})$$

$$T = t_2 - t_0 = 67.800 - 14.500 = 53.300 \text{ } (\mu\text{s})$$

The controller operates with a nominal frequency of 132 kHz, a $\pm 6\%$ tolerance, and an 8 kHz jitter. Under low-load conditions, however, it skips cycles, making the spacing between switching periods unpredictable. To handle this, we will iterate through the data to estimate the number of periods between two successive ON instants.

Let's estimate the max-min frequencies using the tolerance and jitter:

$$\text{Freq}_{cent_{NOM}} = 132 \text{ (kHz, nominal central frequency)}$$

$$\begin{aligned}\text{Freq}_{min} &= \text{Freq}_{cent_{NOM}} \cdot (1 - 0.06) - 8 \\ &= 132 \cdot (1 - 0.06) - 8 \\ &= 116.080 \text{ (kHz, min frequency, jitter and tolerance)}\end{aligned}$$

$$\begin{aligned}\text{Freq}_{max} &= \text{Freq}_{cent_{NOM}} \cdot (1 + 0.06) + 8 \\ &= 132 \cdot (1 + 0.06) + 8 \\ &= 147.920 \text{ (kHz, max frequency, jitter and tolerance)}\end{aligned}$$

Let's iterate through the data to estimate the number of periods between two successive ON instants.

Frequency for 5 periods: 93.81 kHz
 Frequency for 6 periods: 112.57 kHz
 Frequency for 7 periods: 131.33 kHz
 Frequency for 8 periods: 150.09 kHz
 Frequency for 9 periods: 168.86 kHz

Regarding the max min possible frequencies, we can see that 7 periodes is the good value. so we can calculate the duty cycle now:

$$N_{periode} = 7$$

$$\text{Freq} = \frac{N_{periode}}{T \cdot 1 \times 10^{-6}} = \frac{7}{53.300 \cdot 1 \times 10^{-6}} = 131332.083 \text{ (Hz)}$$

$$\text{Freq}_{khz} = \text{Freq} \cdot 1 \times 10^{-3} = 131332.083 \cdot 1 \times 10^{-3} = 131.332 \text{ (kHz)}$$

$$\begin{aligned}\text{Duty}_{cycle} &= 100 \cdot T_{on} \cdot 1 \times 10^{-6} \cdot \text{Freq} \\ &= 100 \cdot 4.300 \cdot 1 \times 10^{-6} \cdot 131332.083 \\ &= 56.473 \text{ (\%)}\end{aligned}$$

The datasheet [1] duty cycle min max typ are 62, 65, 68

Change the transformer airgap

A test using the same design at $V_{in} = 420\text{V}$ was performed, showing that the magnetizing inductance decreases from $1650\mu\text{H}$ at 360V to about $500\mu\text{H}$ at 420V . This occurs because the core becomes saturated: although the TNY-264 limits the current, the limit depends on the current slope, and at 420V the effective current limit exceeds 320mA , leading to saturation.

As a solution to the core saturation, we will add an air gap to increase the current slope, reduce the voltage-time integral, and lower the core's peak flux density.

Below are the test results with this new air gap.

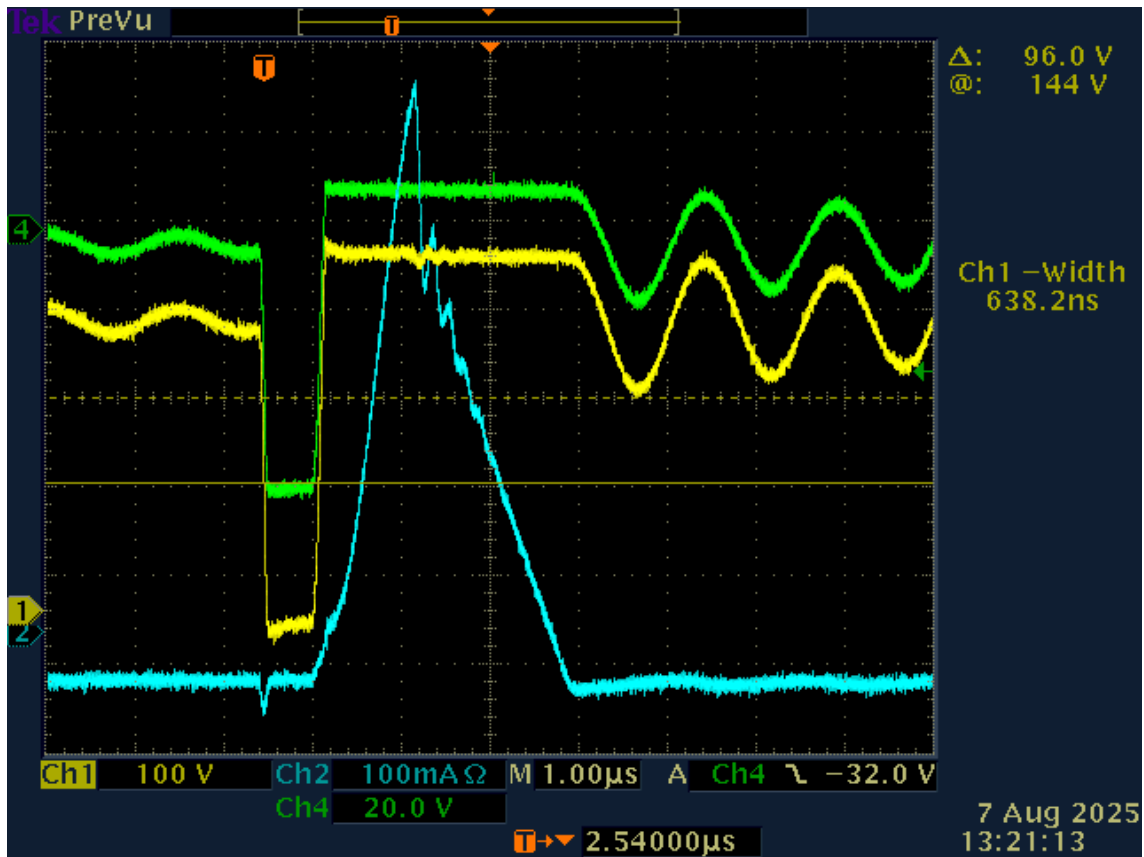


Figure 26: Oscilloscope (screenshot): CH1= Vds, CH2= Is-Transfo, CH4 = Vs-Transfo

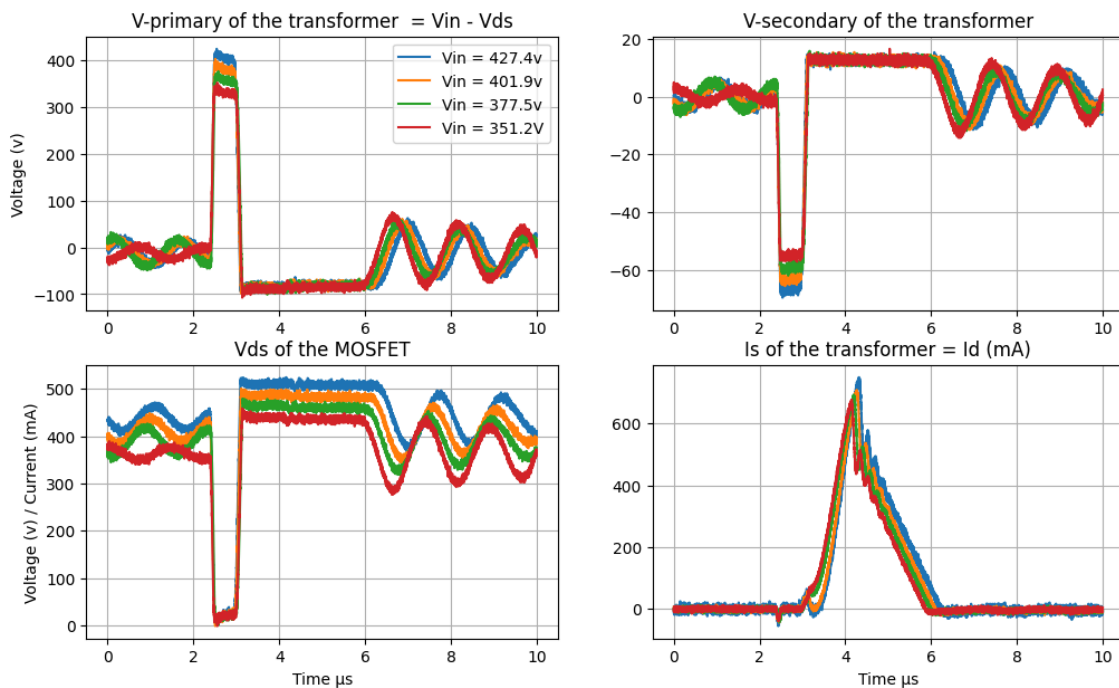


Figure 27: Oscilloscope (data): Pre-processing to synchronize the ON instant

All waveforms remain stable as the input voltage increases, which is a clear indication that the core is no longer saturating.

Since the primary current is not measured (to avoid adding a measurement loop that could increase leakage), we estimate it using the method shown below: applying the voltage drop across $R_{ds} = 44.57$ (previously calculated) and the measured V_{ds} .

We will apply linear regression as a filtering method in each graph.

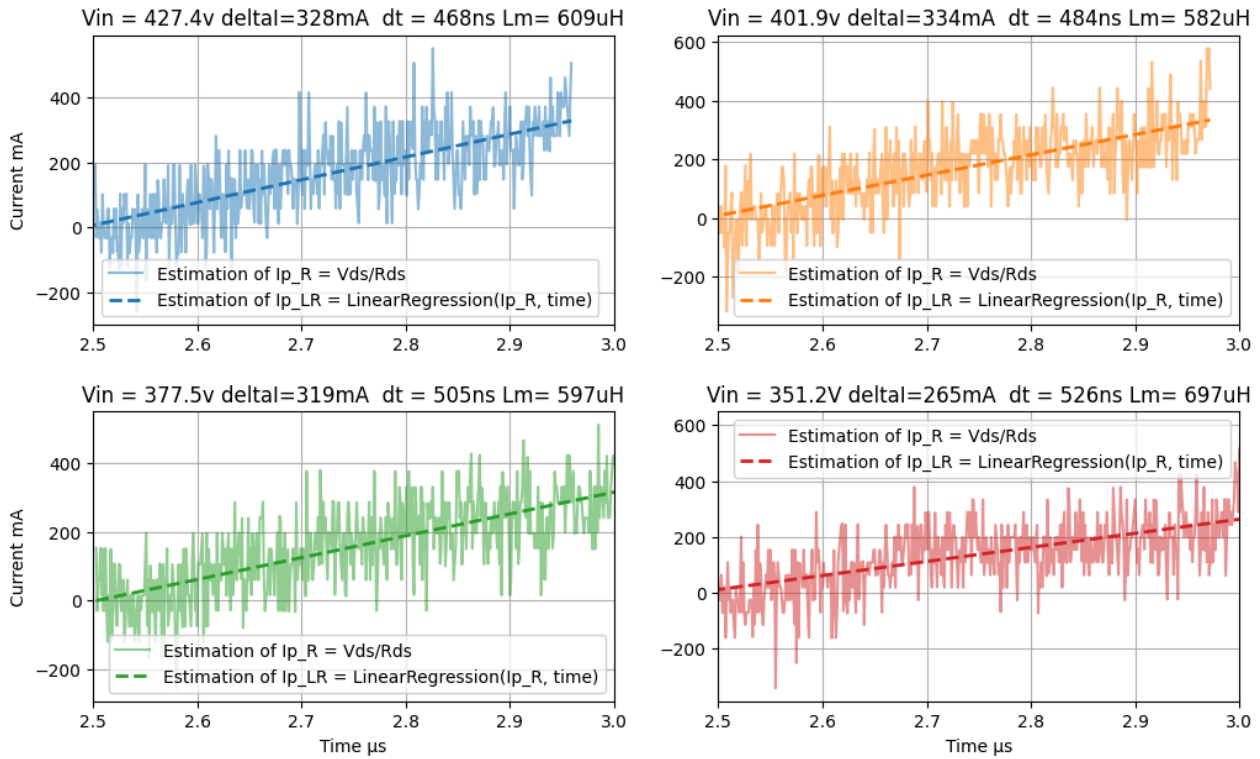


Figure 28: Pre-Processing: The estimation of L_m

The magnetizing-inductance estimation results are shown below. Although L_m decreases, it stays safely outside the core-saturation region.

V_{in}	dt_{ns}	di_{mA}	$L_{m_{uH}}$
351	525	264	697
377	505	319	596
401	484	334	581
427	468	328	609

Zoom on the diode ON phase:

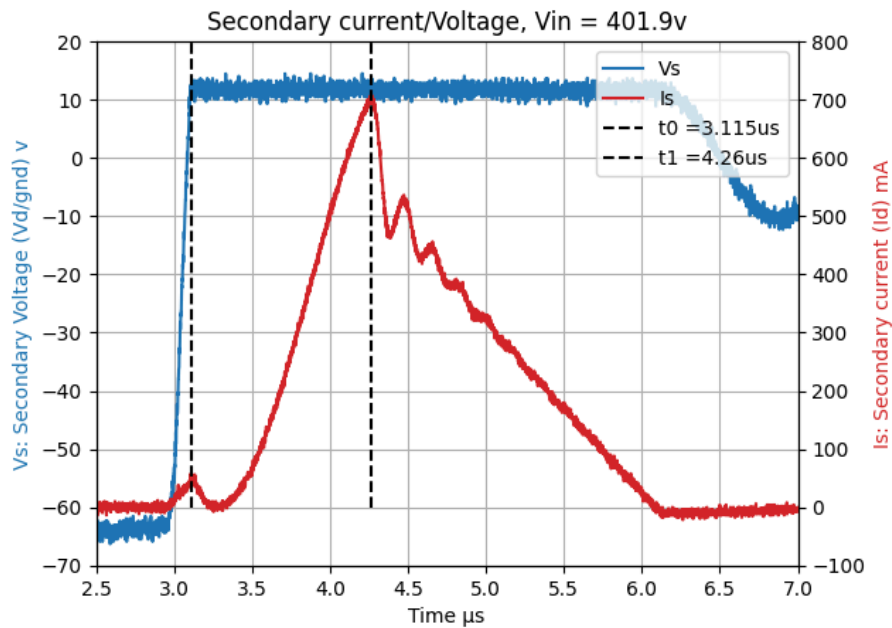


Figure 29: TEST

Understand the flyback waveforms

To understand the flyback waveforms, we use the simplified model shown below.

It simulates the moment when the MOSFET turns OFF:

- We observe the characteristic voltage Vsn0: the clamp voltage, then fast ringing caused by Llk and Coss, followed by slower ringing between Lm and Coss.
- The secondary current is also visible: it ramps up immediately after the MOSFET turns off, then decreases with superimposed fast ringing

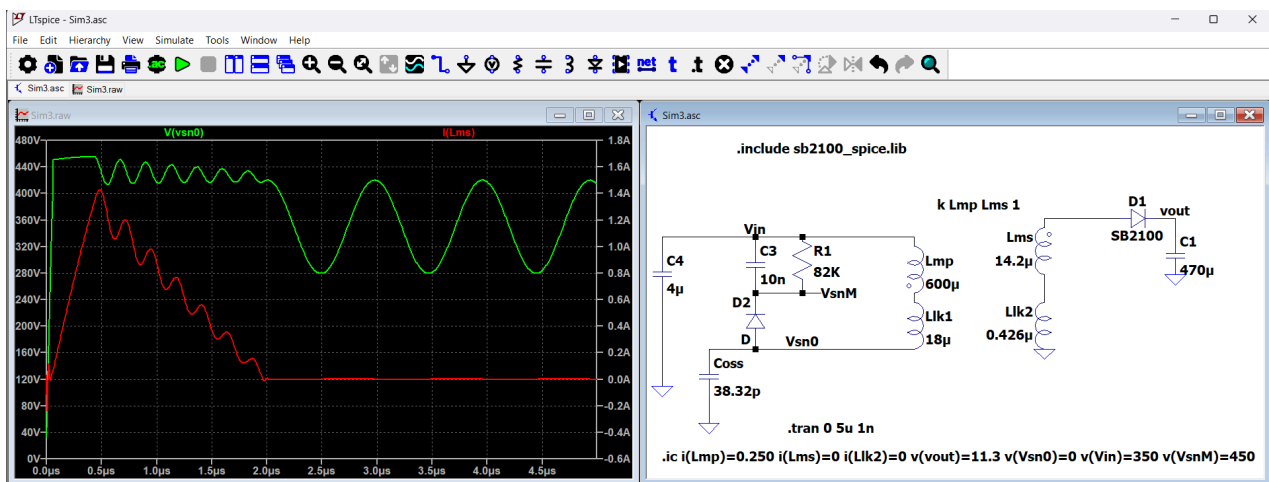


Figure 30: Simulation to understand the flyback waveforms

We can compare the simulation with the measurement results, even though the MOSFET is not fully modeled (it is represented only by its Coss).

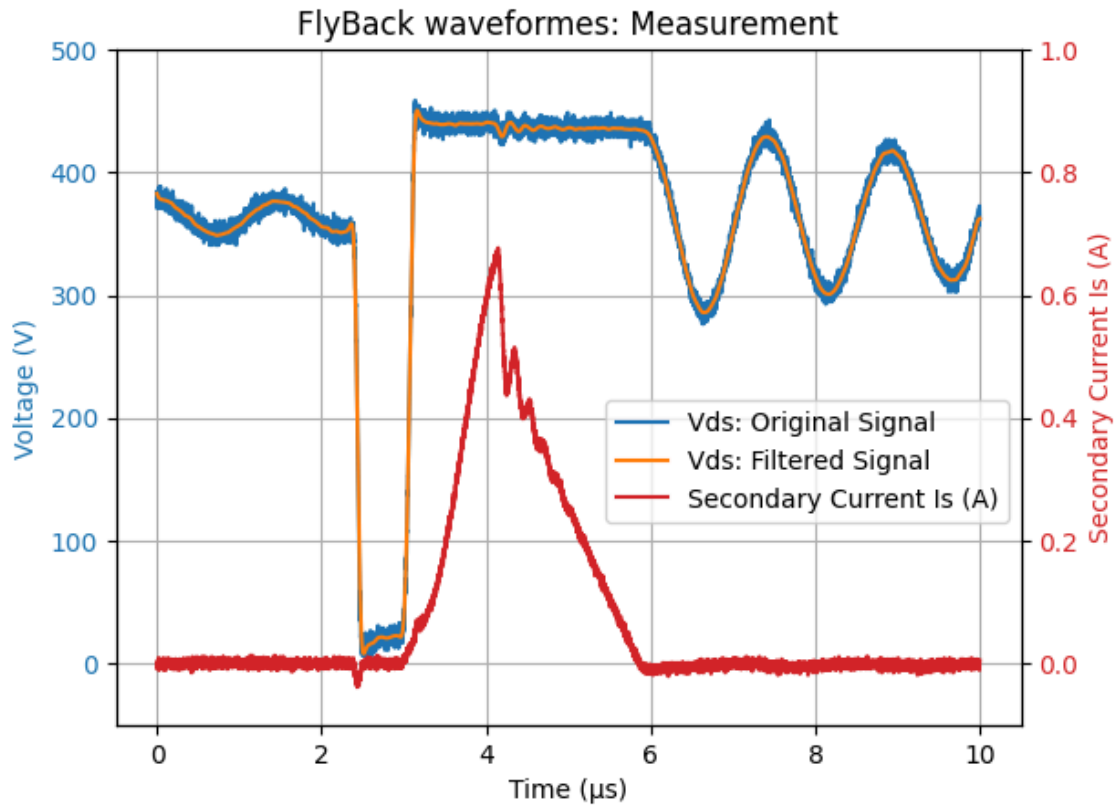


Figure 31: Flyback waveform measurements

We can model the MOSFET as a perfect switch with a parallel Coss capacitor and a 40Ω series resistor, and we obtain the same behaviours. See the simulation below:

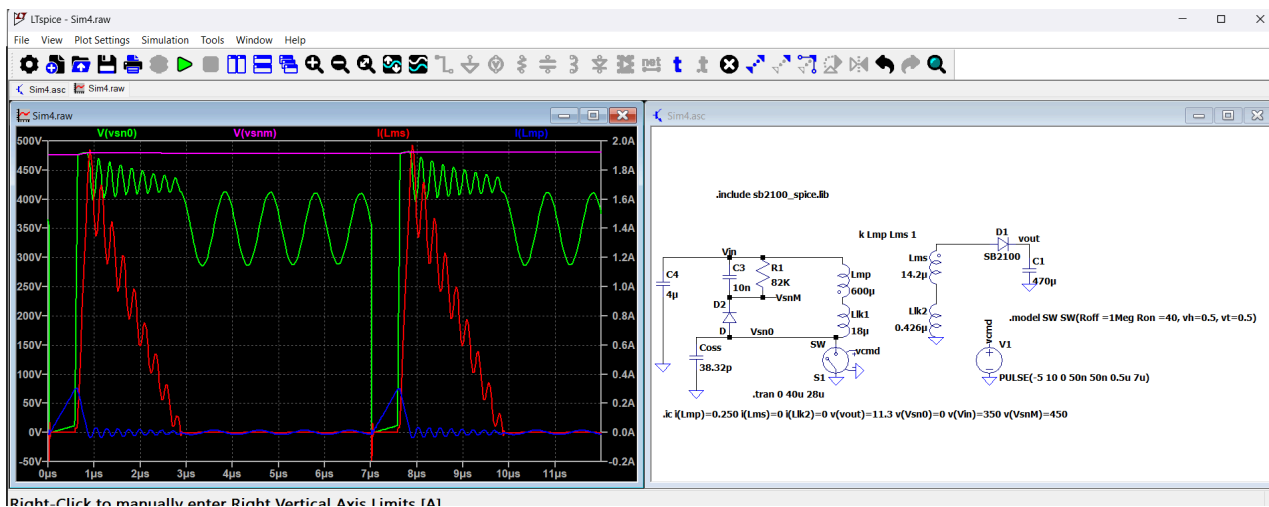


Figure 32: Simulation 2 to understand the flyback waveforms

You can find all Itspice simulations of this flyback at the [link](#).

Vout ramp

We measured the Vout ramp, which is shown in the figure below:

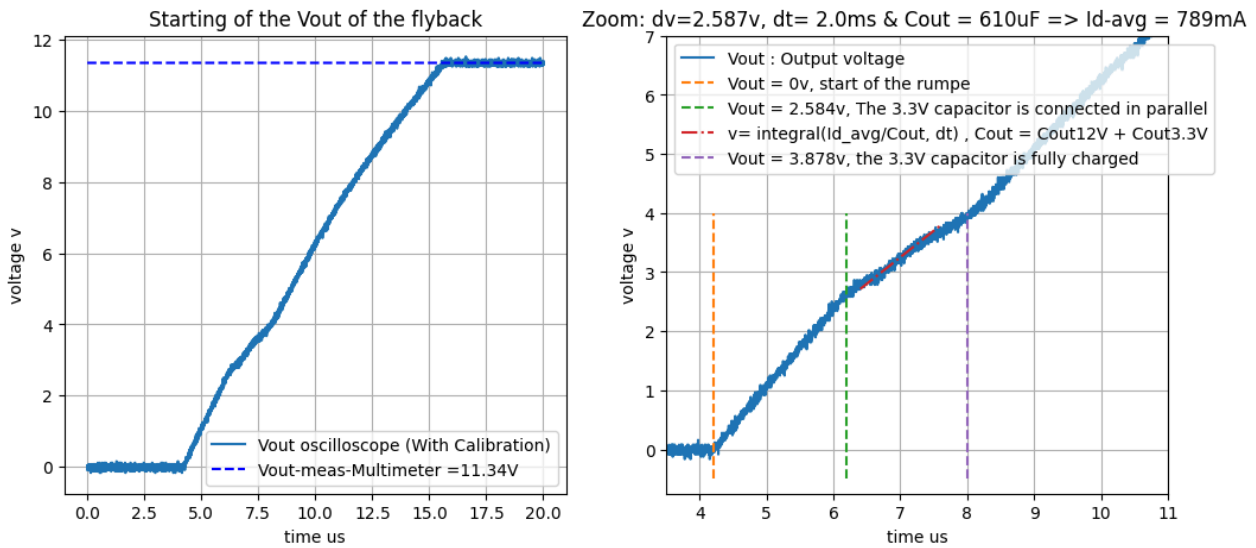


Figure 33: Vout ramp

The ramp is composed of three segments:

- A fast slope from 0 to 2.584 V, explained by the flyback output capacitor (470 μ F).
- A slower slope from 2.584 V to 3.878 V, caused by the flyback and buck capacitors being in parallel. Initially, the buck capacitor is charged, so the buck acts like a diode. Once Vout exceeds the capacitor's initial voltage, the diode conducts and the two capacitors are effectively in parallel.
- Then a second fast slope from 3.878 V, similar to the first. In this region, the buck starts converting and isolates the flyback capacitor from the buck capacitor.

Secondary short-circuit

In the figure below, there are three plots:

- Left: raw measurements of primary voltage and current. From these, we calculate $T_{on} = 400$ ns. With a switching frequency of 132 kHz, this results in a duty cycle of **5.28%**, which matches the minimum duty cycle of the TNY-264 controller (**5.6%**).
- Middle: flux calculation and comparison with an ideal inductance equal to the leakage inductance.
- Right: by comparing this plot with the middle one, we see that the primary inductance nearly matches the previously estimated leakage inductance (Llk1).

Conclusion:

The controller and circuit behave exactly as expected, and the restart time of about 800 ms is very close to the controller's specification.

TNY264 enters default protection due to a shorted output transformer winding

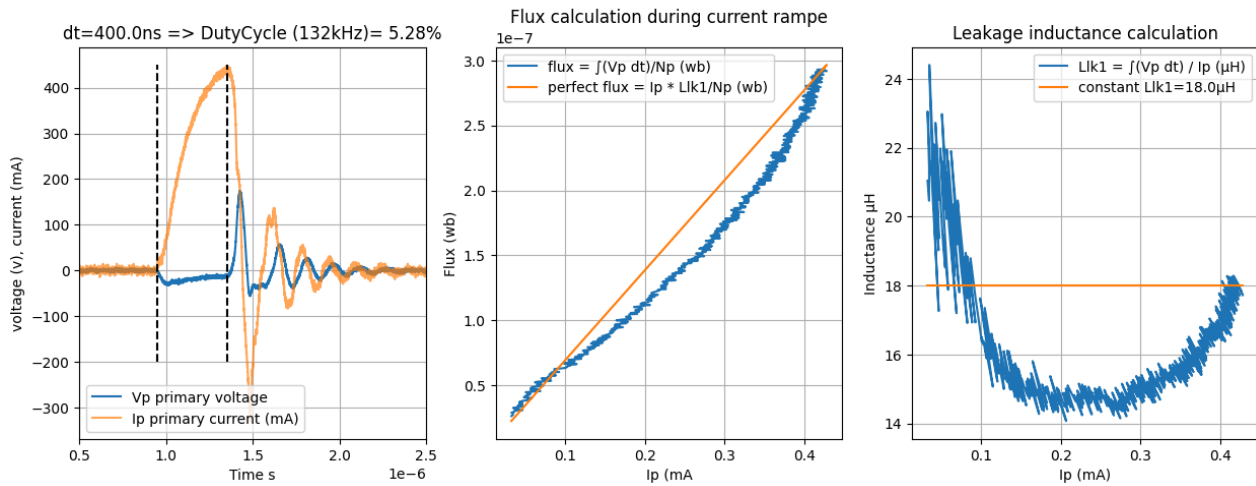


Figure 34: Secondary short-circuit

Rectifier diode modification

As mentioned in the table in the **Modifications** section, see the screenshot of the table below.

Change	Before	After	Why	Impact
Output diode	MBRS340T3G	SB2100	VRRM = 40 V (old) vs 100 V (new) VRRM of new design ≈ 62 V (peak)	Not SMD PCB adaptation

In this section, we will examine the waveforms before and after modifying this diode.

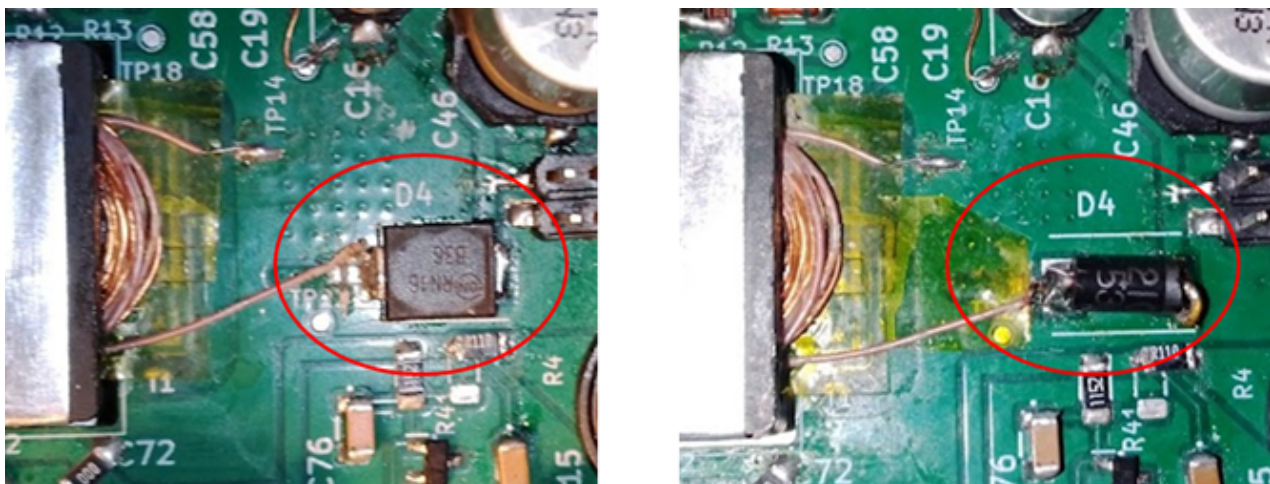


Figure 35: Output diode: Before (right) and after (left) modification

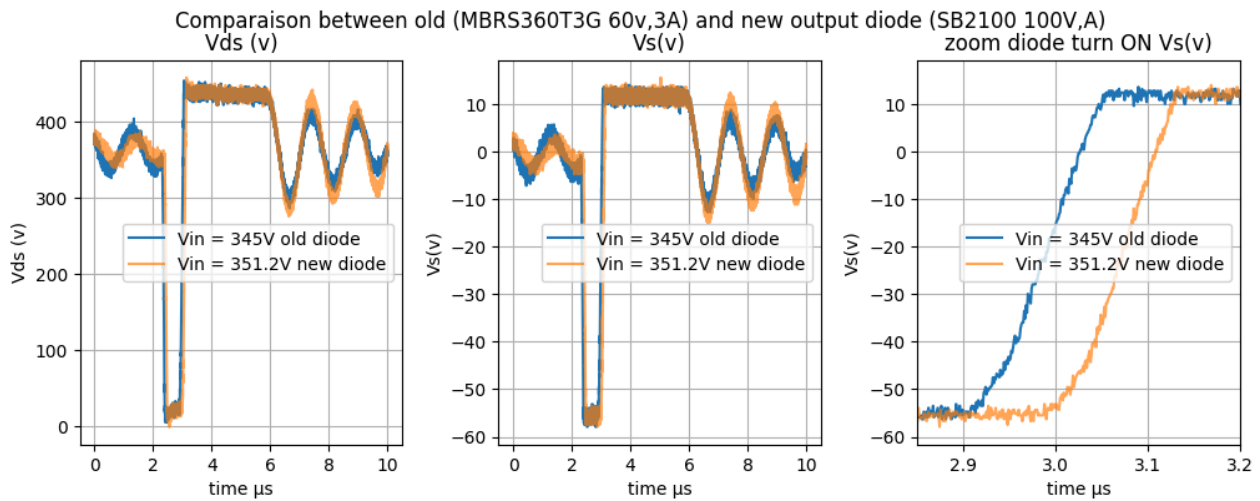


Figure 36: Comparison before and after diode modification

The new output diode behaves almost the same as the previous one, but with greater VRRM margin (see above).

References

[1] Datasheets

TNY-264

<https://www.farnell.com/datasheets/98510.pdf>

ELP 22/6/16 with I 22/2.5/16 N87

https://www.tdk-electronics.tdk.com/inf/80/db/fer/elp_22_6_16.pdf

TNY280

https://www.power.com/sites/default/files/documents/tinyswitch-iii_family_datasheet.pdf

TNY-264

<https://www.farnell.com/datasheets/98510.pdf>

MBRS340T3G

<https://www.onsemi.com/download/data-sheet/pdf/mbrs340t3-d.pdf>

US1J

<https://diotec.com/request/datasheet/us1a.pdf>

US2MA

<https://www.onsemi.com/download/data-sheet/pdf/us2aa-d.pdf>

Output capacitor:EEVFK1V471Q 470uF 35V 20%

<https://industrial.panasonic.com/cdbs/www-data/pdf/RDE0000/ABA0000C1181.pdf>

[3] Fairchild: Application Note AN-4147

[4] TDK, Ferrites and accessories, E cores General information

[5] CIRCUIT DESIGNER'S NOTEBOOK, page 5, see the example